

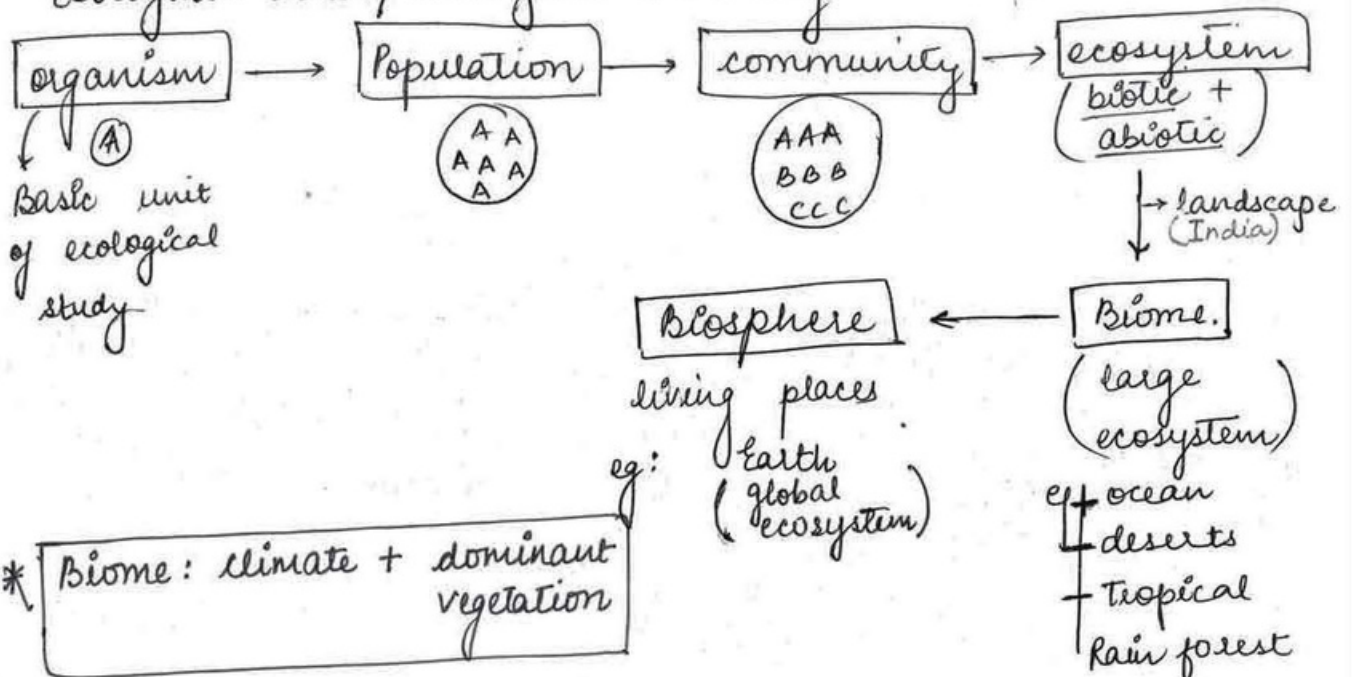
Ecology = Father : Reiter

ecology $\leftarrow$ oikos $\Rightarrow$ home/habitat
 $\leftarrow$ logos $\Rightarrow$ study

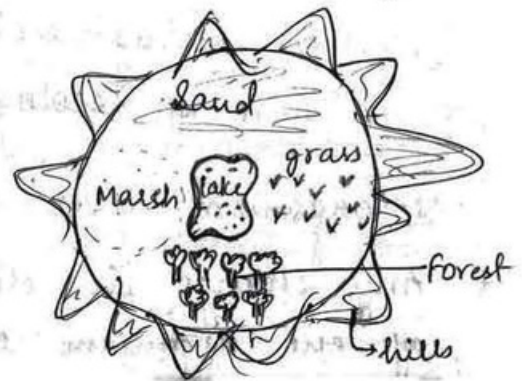
Organism $\xrightleftharpoons{\text{interaction}}$ environment
 $\updownarrow$
 organism

Biotic $\rightarrow$ Plants
 animals
 microbes
 Abiotic $\rightarrow$ light
 temp
 water
 soil

Ecological order / Ecological hierarchy



landscape:
 group of different ecosystem
 surrounded by natural boundary.
 eg: India



4 Basic level of ecological order:

- Organism
- Population
- community
- ↔
- Biome
- ↔

~~Single~~

Branches (on organism level)

Autecology

Synecology

~~Single~~ single species $\rightleftharpoons$ envir.

group of species $\rightleftharpoons$ envir.

Unit of study single species/
single member

Unit of study:
community

i.e. species ecology
ecology of individual

i.e. community ecology

eg: effects of global warming
on human

effects of global
warming on others

Organism:

Smallest living unit

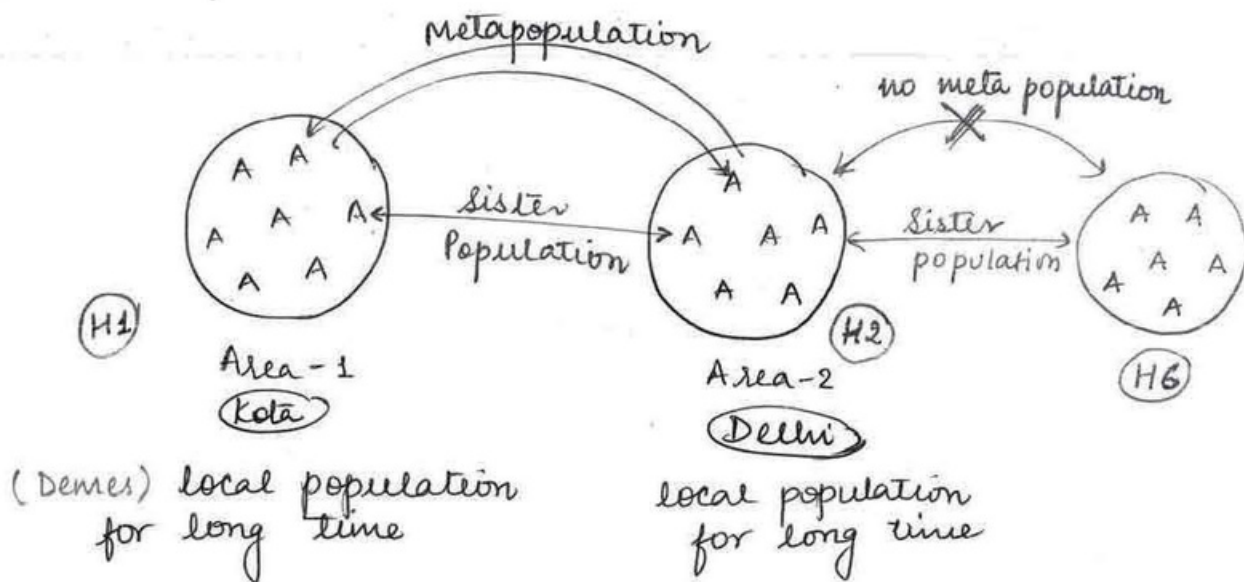
Basic unit of ecological study (~~DEPT~~)

At organism level ecology $\Rightarrow$ Physiological ecology

* Any change in climate will affect the physiology of the organism like sweating in summer and shivering in winter

Population

group of similar individual (species) live in a particular area

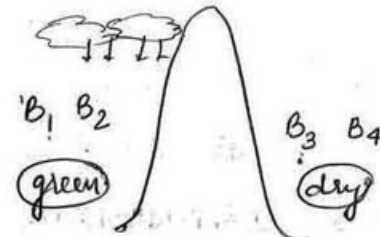


Ecotype →

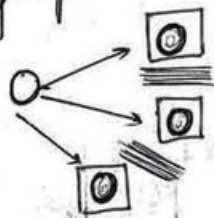
Similar species

Different environmental condition

Genetic Variation ⇒ Present **
(no fundamental changes)



Allopatric species → species that are found in different geographical area

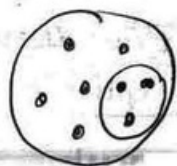


forms different species

Sympatric species

same geographical area

forms different species



Ecades / Ecophens :

Environment → Different

genetic variation → absent

external variation → present (Reversible)

Phenotypic Plasticity

eg: Money plant, Humans (well Indian n England)

Species

Endemic species

found only in geographical area

- less number

∴ more chances to extinct

due to formation of CONTINENTAL SEPERATION

eg: Kangaroo in australia

g Metasequoia in china valley

- Kiwi in new zealand

All endemic species are rare species

* Key stone species

number ↓

Biomass ↓

Fresh wt

Dry weight

(BOMB CALORE METER)

Influence ↑↑

maintains structure and balance

* eg: Tropical deciduous forest

f. (leaves fall down in summer)

Fig tree

(remains green)

eg: Kangaroo Rat in desert

Burrow → Reptiles prevents

eg: Lion in forest

critical link species (Mediator)

Mediator species that are helpful for survival of other species (due to some vital functions)

eg: Fungi in mycozzhiza

eg: grass → Rabbit → wolf

∴ wolf can't eat grass directly.

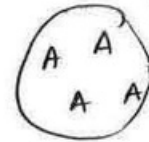
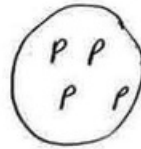
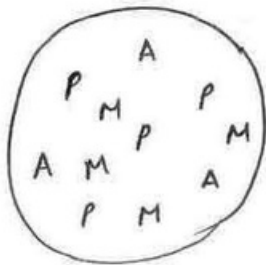
Key Stone

- > number ↓ i.e (less)
- > if removed than eco-system will destroy after some time

Critical link

- number ↑ i.e (more)
- no such function

Biotic community : Plant community + Animal community + Microbial community



Characteristics :

1. Species diversity : (structural)
no of species
≠ no of organism (individual)

* Factors affecting species diversity

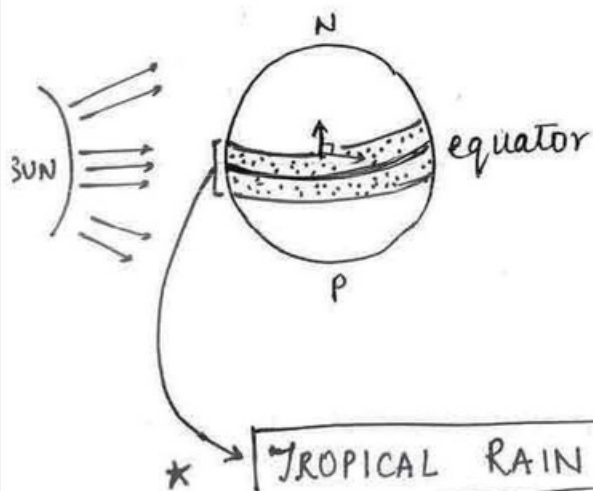
① CLIMATE

a) Temperature : main factor that effect distribution and diversity of plants

b) water : Second factor

c) light : productivity eg: Photosynthesis

② TYPE OF AREA :



latitude: distance from equator ↑
Temp ↓ diversity ↓

Altitude: height above sea surface ↑
Temp ↓ diversity ↓

* **TROPICAL RAIN FOREST** : Max. diversity

③ TYPE OF SOIL: soil related factors [Edaphic factors]

④ SIZE OF AREA: area ↑ diversity ↑ (but upto a limit)

Q) A, B, C (Birds)
X (mammals)
I (Insects)

Area : 1

A	A	C
B	A	A

3 sp

✓ Related (birds)
✓ uneven (4, 1, 1)

[less diversity]

Area : 2

A	B	C
A	B	C

3 sp

✓ Related (birds)
✓ even (2, 2, 2)

[more diversity]

Area : 3

A	X	I
A	X	I

3 sp

✓ unrelated
✓ even (2, 2, 2)

Most diverse

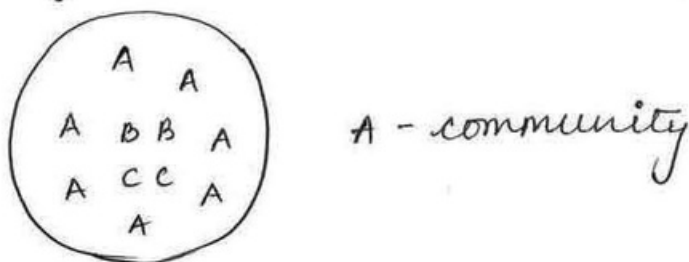
Diversity $\pm$ species richness ($\Rightarrow$)
 species evenness ($\hookrightarrow$)

2.

structural $\Rightarrow$ physical app. *
 functional $\Rightarrow$ functions

2. Dominance (structural)

highest no of organism of a species in community

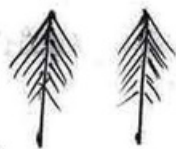


eg: Pinus community: ~~Aravali hills~~ Himalaya
 Mangroove community: Sundarbans (West Bengal)
 Prosopis community: Aravali hills

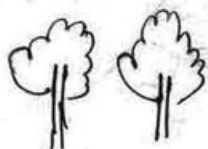
sit in helicopter and visualise :P

* Physiognomy: External appearance or looking of plant community is called. physiognomy, it is determined by dominant plant community (no. + biomass)

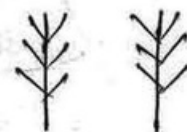
eg:



coniferous forest



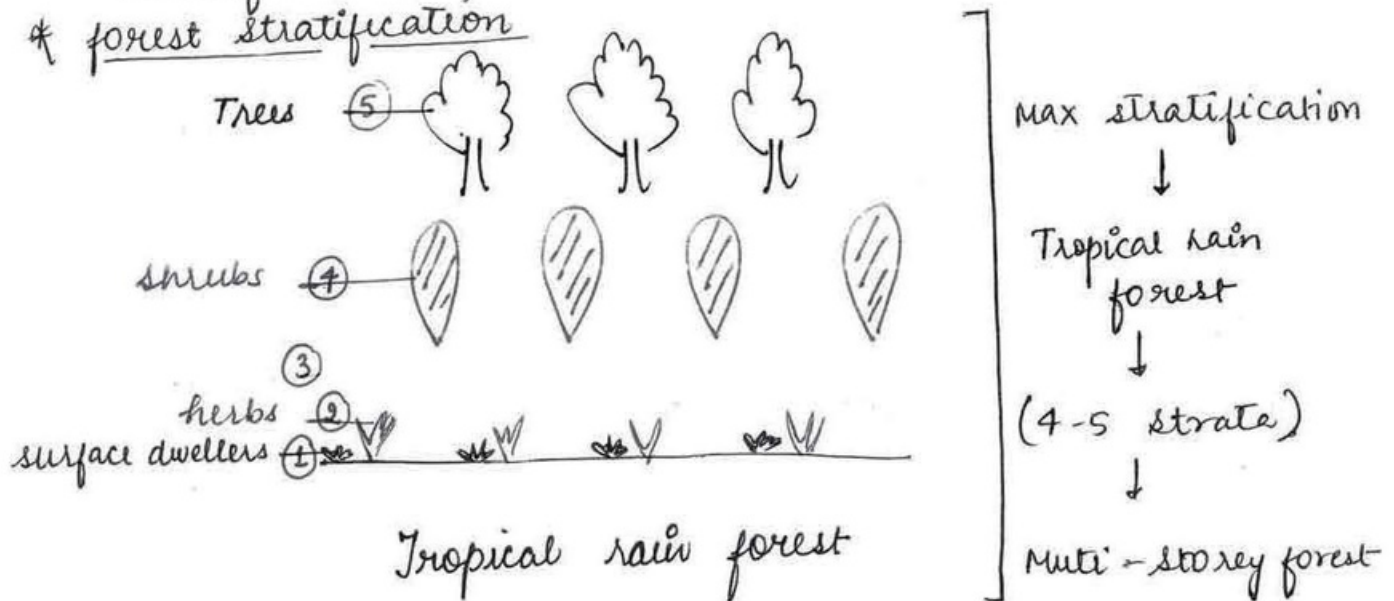
Tropical rain forest



Tropical deciduous forest

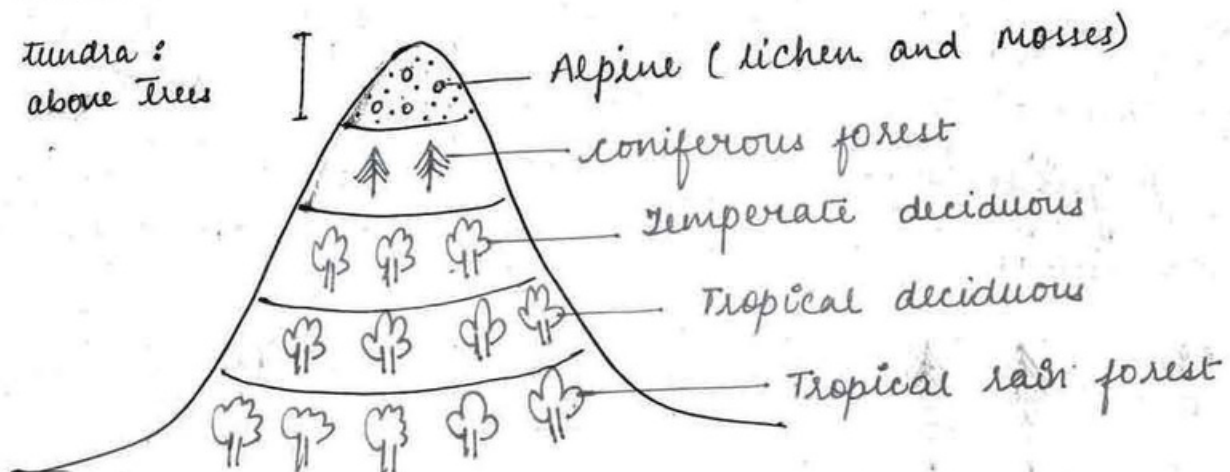
Vertical layering of plant vegetation according to their growth forms is called stratification

* forest stratification



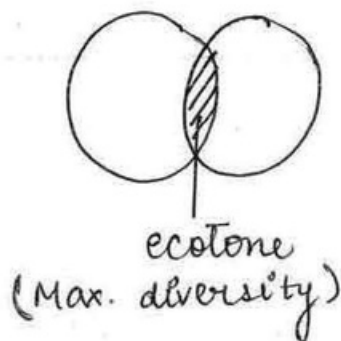
Q) why stratification occurs in rain forest?
According to need of light or efficient use of resources.

* Mountain stratification $\Rightarrow$ horizontal stratification.



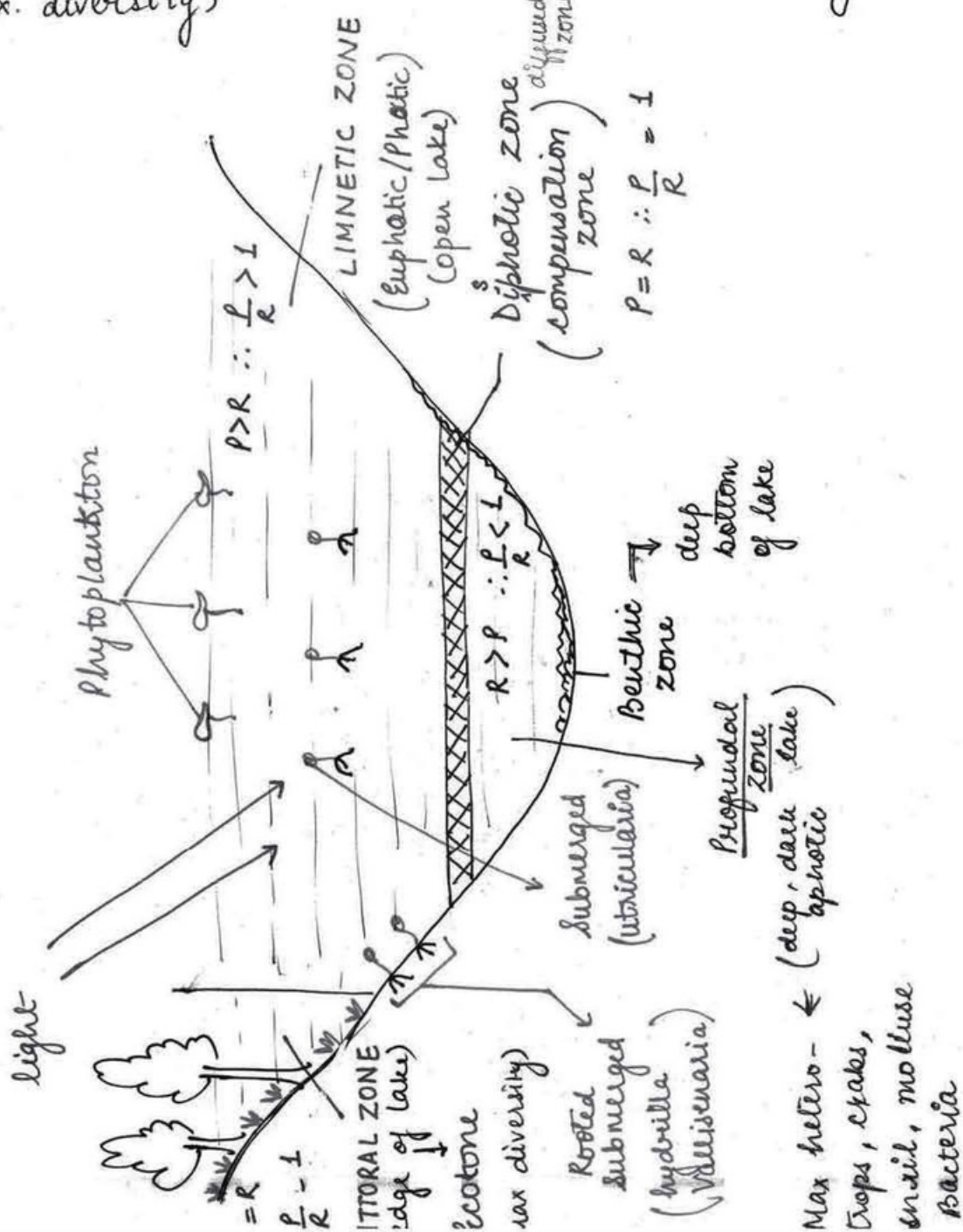
12 stratification is based on temp due to $\uparrow$ in altitude

* Lake stratification $\Rightarrow$ horizontal and vertical stratification.
(light basis)



$\Rightarrow$ Transition area b/w 2 community

$\Rightarrow$ One community modifies to another community



Plankton: (free-swimmers)

They do not have their own movement

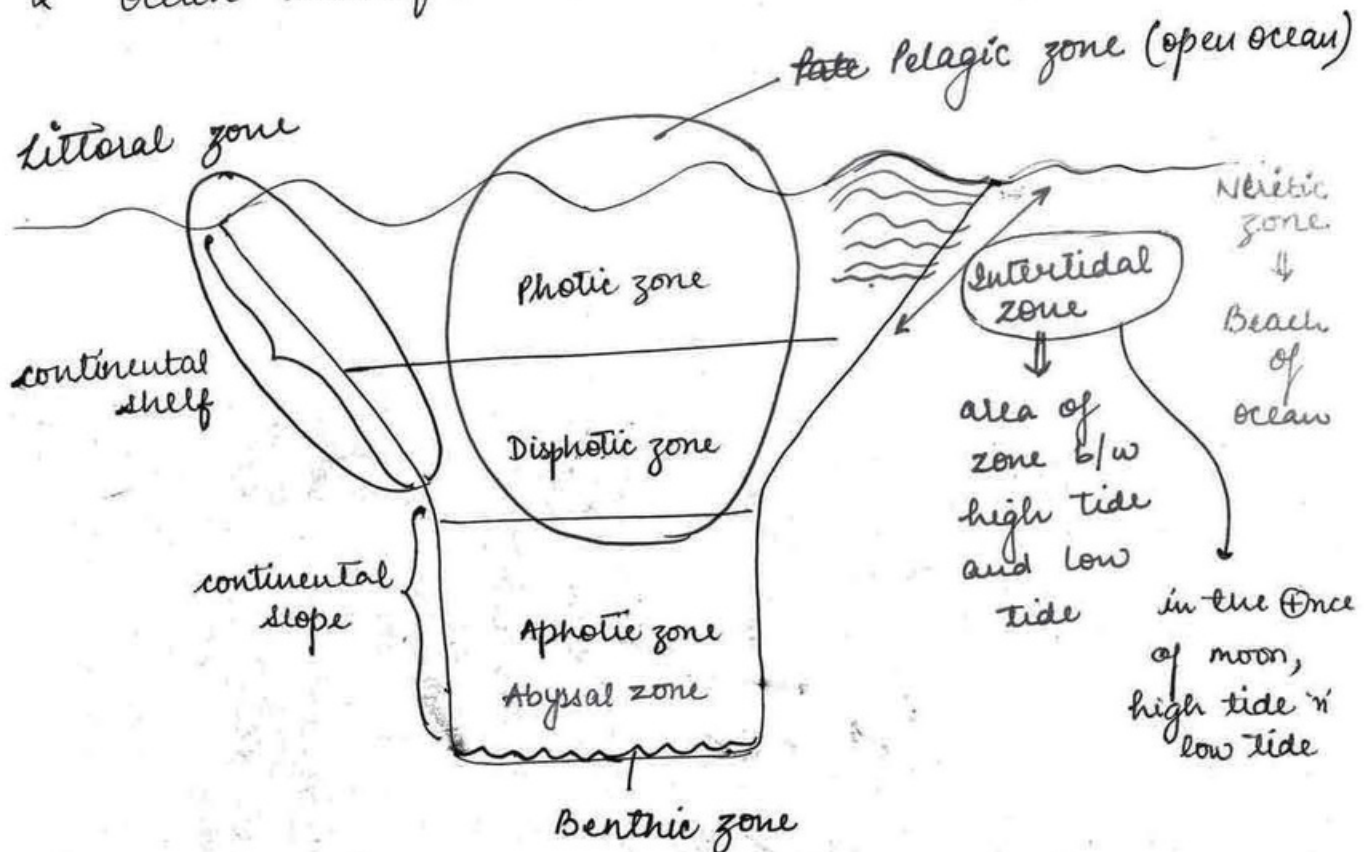
They move with the waves of water

eg: diatoms

Vegetons: They have well developed flagella for their movement

eg: diatoms

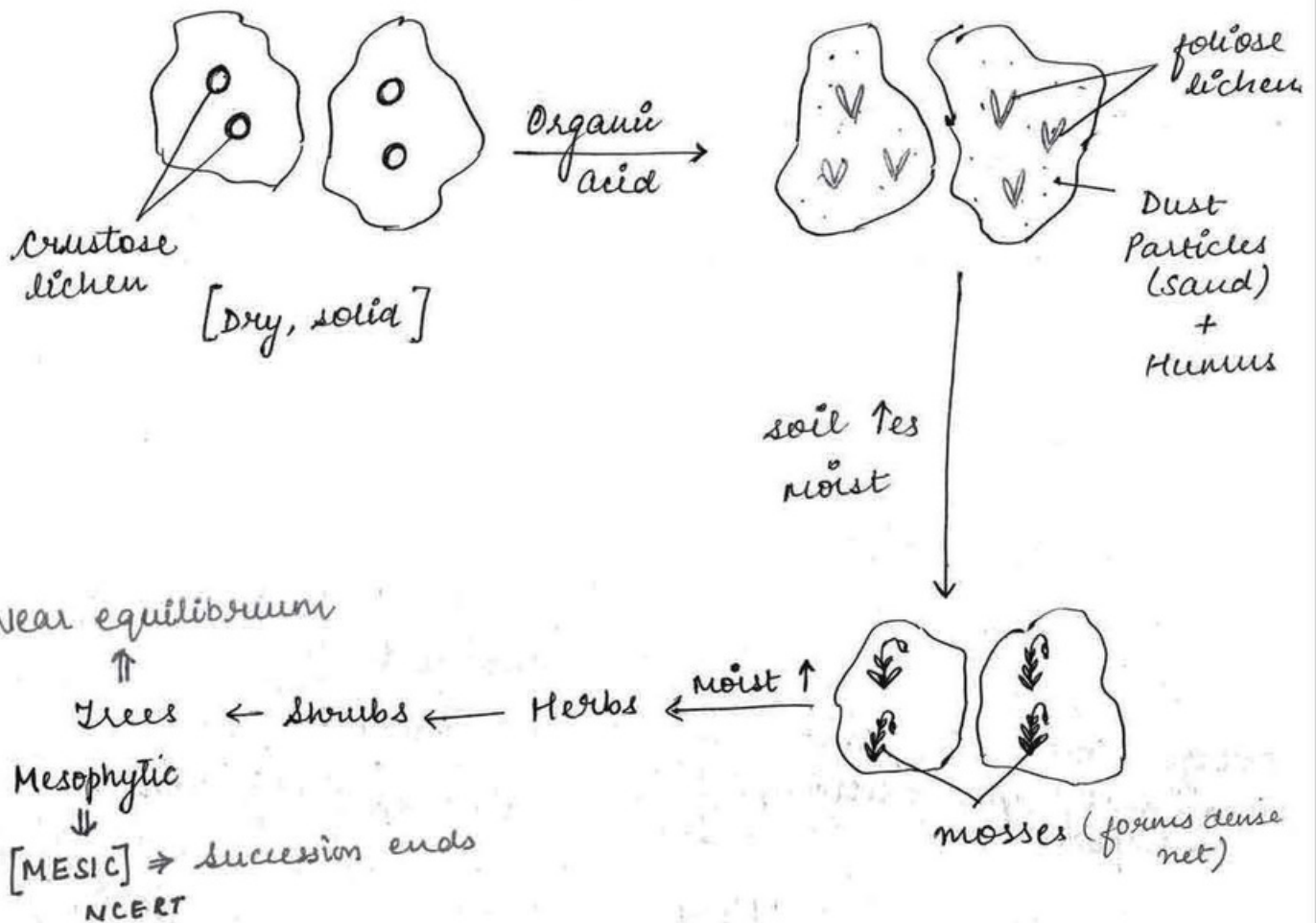
* Ocean stratification:



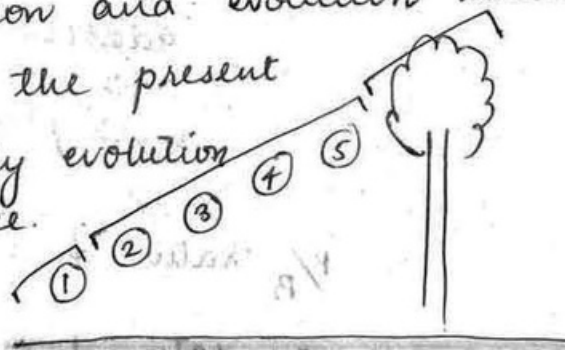
Succession

- * Biotic community is not stable.
- * It is a predictable process

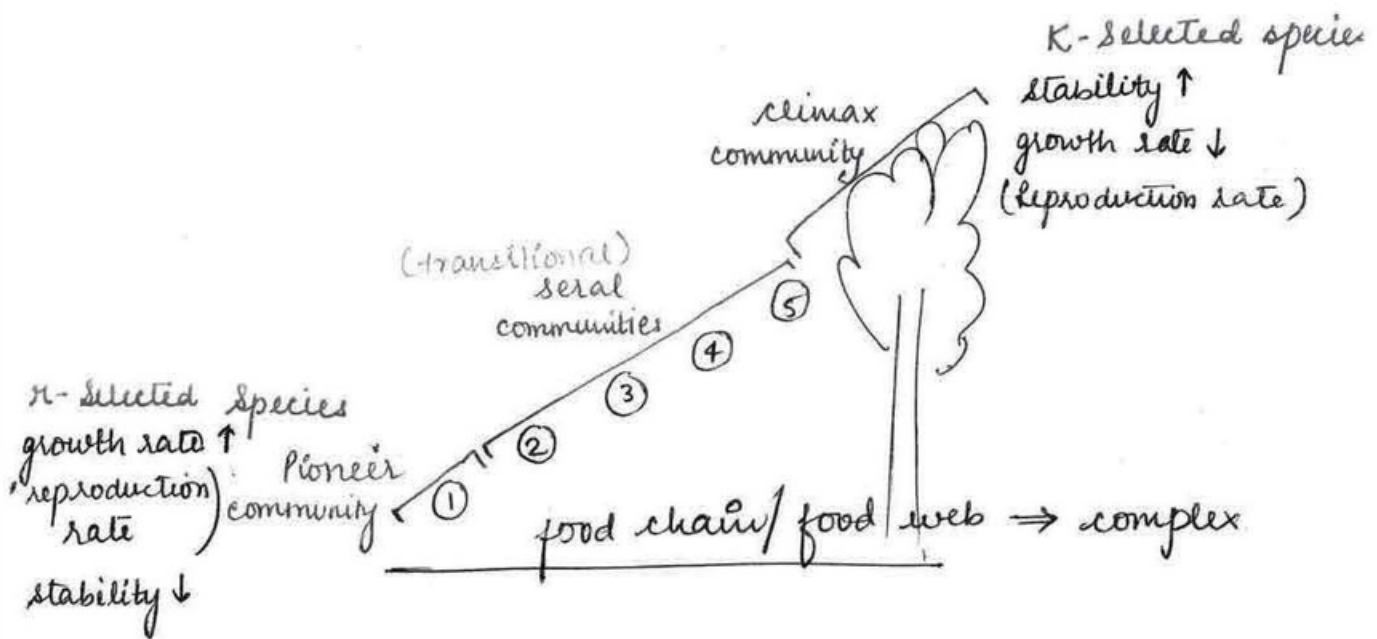
Lithosere → succession on rock



* Long back ago after the form of earth, succession and evolution were parallel process. but at the present time only evolution takes place.



P.T.O



short lived → long lived
 small size → large size
 Biomass ↑

∴ consumption ↑ by other organism
 net yield ↓
 Energy stored in plant
 community after heterotrophic
 consumption

Humus ↑
 diversity ↑
 Niche specialization ↑

* Net community Productivity ↓
 (NCP)

Niche

Functional
~~role~~ role
 ↓
 work

Occupational
 address
 ↓
 work place

P/B ratio ↓

P/R ratio ↓

	<u>Seral community</u>	<u>complex community</u>
i) Organism size	small	large
ii) Niche	generalised	specialized
iii) community organisation	simple	complex
iv) Food chain	simple	complex
v) Food web	low	high
vi) Nutrient conservation (on decomposition of 1 organism)	low low ☹️	high
vii) Biomass	low	high
viii) Humus	low	high
ix) Diversity	low	high
x) P/B ratio	high 🌍	low
xi) P/R ratio	$P > R$	$P = R$



oxygen formed = O_2 consumed

R = Respiration.

P = Photosynthesis, B = Biomass,

causes of succession:

- i) Biotic : living cause / **Autogenic** (makes unfavourable environment for its survival)
- ii) Physiography : Nature / **Allogenic** (becomes unfavourable due to action of nature)
i.e. Natural calamities

Types of succession:

i) Primary:

Rock $\longrightarrow$ weathering $\longrightarrow$ soil (humus)

$\downarrow$

trees $\leftarrow$ shrubs $\leftarrow$ herbs $\leftarrow$ mosses

(Meric)

eg: Volcanic lava ~~#~~, igneous rock

sand dunes ~~#~~, newly formed ~~the~~ pond

$\gg$ slow process (in 1000's of years)

ii) Secondary:

Forest $\rightarrow$ fire $\rightarrow$ Pteridophytes $\rightarrow$ shrubs $\rightarrow$ tree
(Rhizome)

$\gg$ comparatively fast process (in 50-100 years)

~~the~~ eg: Deforested area, forest fire

① Xerosere
(Dry area)
 $\begin{matrix} \nearrow \\ \searrow \end{matrix}$
XERACH $\begin{matrix} \nearrow \text{Lithosere (Rock)} \\ \searrow \text{Psamosere (sand)} \end{matrix}$

② Hydrosere
 $\begin{matrix} \nearrow \\ \searrow \end{matrix}$
HYDRACH
(fresh water
succession)

③ Oxalosere : Acidic water succession

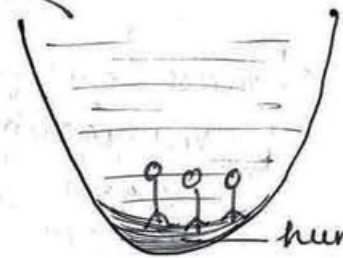
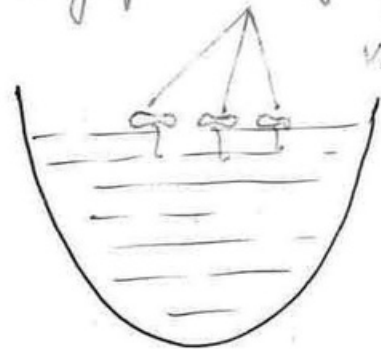
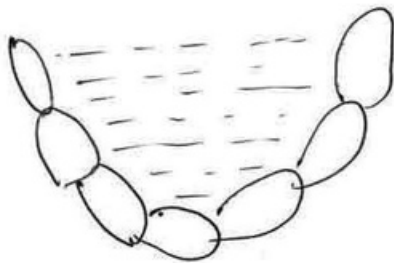
④ Halosere : Salty water succession

* Serula : microbial succession on decomposed matter

eg: saprotrophs $\left\{ \begin{array}{l} \text{saprophyte (an plant decay)} \\ \text{saprozoic (an animal decay)} \end{array} \right.$

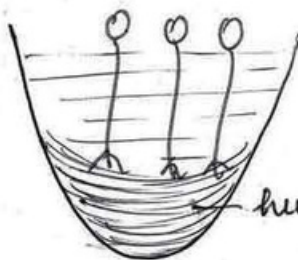
HYDROSERE / HYDRACH

Phytoplankton eg: ^{diatoms} Hydrilla, BGA, Volvox



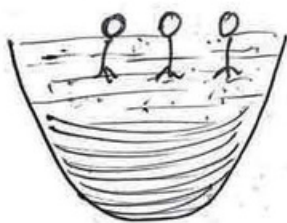
humus

Rooted submerged
eg: Hydrilla, Vallisneria

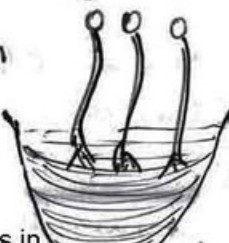


humus

Rooted floating Plant
eg: Nymphaea, Nelumbo, Trapa



free floating
eg: Eichhornia, Pistia



AMPHIBIOUS
STAGE

(woody)
Reed swamp
eg: Ranunculus, Sagittaria



MARSH
Marsh Meadow

eg: Sarcocolla



Scrub
or
Shrub

eg: Lantana



Trees
eg: Oak, Salix

Hydrosere is faster than Xerosere



Pond succession is from margin to centre.

Lithosere

Pioneer : lichen

climax : Tree
(Mesophyte)

Xeric $\rightarrow$ Mesic
slow process
uniform

Hydrosere

Pioneer : Phytoplankton

climax : Tree
(Mesophyte)

Hydric $\rightarrow$ Mesic
fast process

** Margin to centre

Species Interaction

Intra-specific Interaction

$A \leftrightarrow A$
 $B \leftrightarrow B$

$\oplus$ ve

Mating Behaviour
Aggregation
communication

Altruism (Sacrificial behaviour)

+ Honey bee

$\ominus$ ve

food,
shelter,
space

Inter-specific Interaction

$A \leftrightarrow B$

$\oplus$ ve

Mutualism (+/+)
Commensalism (+/0)
Proto Co-operation (+/+)

$\ominus$ ve

Competition (-/-)
Amensalism (-/0)
Exploitation (+/-)
Parasitism

MUTUALISM / Obligate Mutualism / Symbiosis (+/+)

eg: Lichen $\Rightarrow$ Algae + Fungus
(slave) (Master) $\Rightarrow$ Helotism

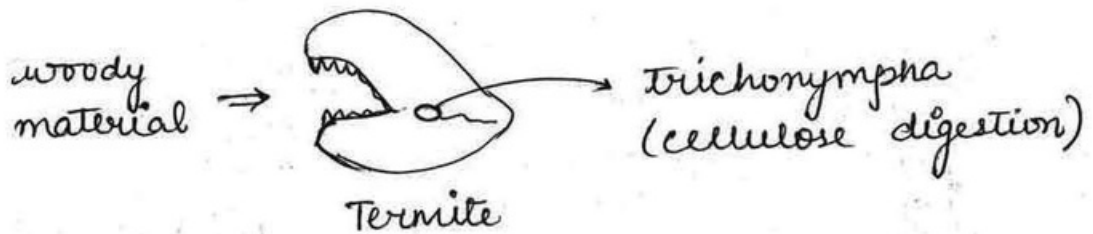
$\downarrow$
food
for^m
(wife)

$\downarrow$
Protection
+
mineral absorption
(husband)



eg: Mycorrhiza $\Rightarrow$ fungi + higher plant root
 $\downarrow$ $\downarrow$
Protection Nutrition
+
mineral absorption

eg: Termite and ~~tree~~ trichonympha



In above three example, they are physically associated

eg: Yucca plant and Pronuba
(Pollination) (oviposition-site)

$\downarrow$ seeds
egg laying site $\rightarrow$ larva $\rightarrow$ adult

Co-evolution: Saath Saath change hona

Co-existence: Saath Saath jeena



NCERT eg: Fig tree and wasp (नरैया)

egg laying site
↓

↓

reception

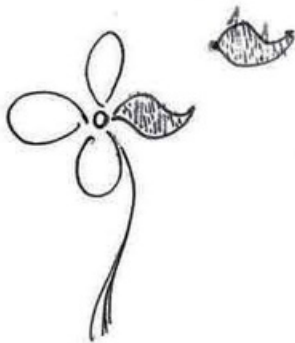
just
for

NCERT १८

шар (сера аица)

↓

Sexual deceit



COMMENSALISM (+/0)

eg: Lianas / woody climbers / woody vines :



↓
light

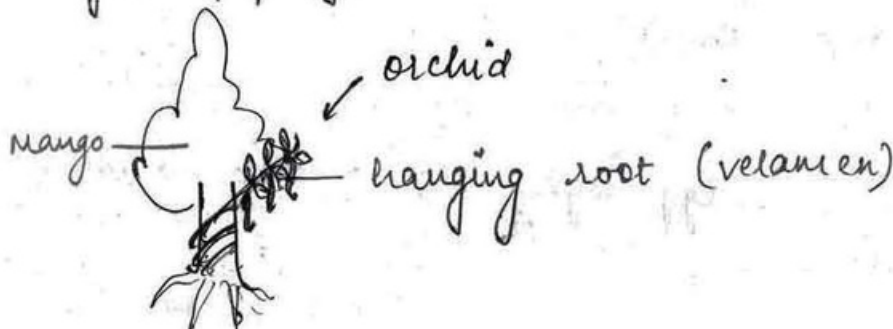
eg: Bauhinia

mostly found in
dense forest

dense forest

tropical rain
forest

eg: epiphyte



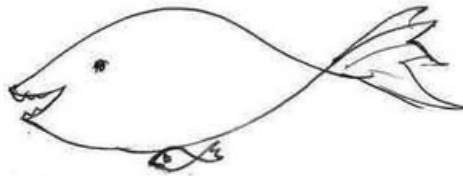
大學

light and
moisture

eg: epizones:

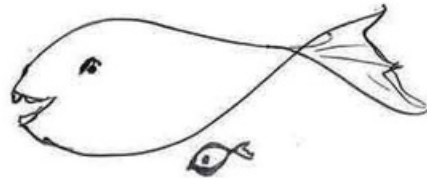
a) sucker fish and shark

- Transportation
- Protection
- food



b) Pilot fish and shark

- Protection
- food



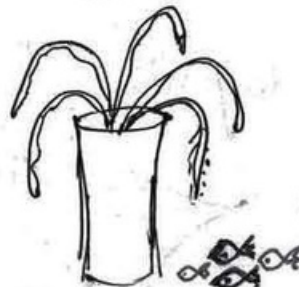
c) E. coli and Human intestine

- shelter
- vit B₁₂ (minor)
- Nutrition

E. coli forms vit. B₁₂ in humans but in very less amount which doesn't fulfill human require (∴ it is ≠ mutualism)

✓ d) Clown fish and sea anemone

- Protection



(N) Barnacles and whale $\Rightarrow$ herbivorous
 (Crustaceans)
 $\downarrow$
 $\begin{cases} \rightarrow \text{Transportation} \\ \rightarrow \text{Protection} \end{cases}$ $\downarrow$ filter feeding

(N) cattle egret birds and cattles
 $\downarrow$ while
 $\begin{cases} \rightarrow \text{insect comes outside} \\ \leftarrow \begin{cases} \text{locomotion and grazing} \end{cases} \end{cases}$



PROTO-CO-OPERATION / (+/+)
 Non-obligatory mutualism /
 Facultative mutualism (Tu nhi toh koi aur hi sahi)

eg: Tuck bird and Rhinoceros
 $\begin{cases} \rightarrow \text{eat from rhinoceros waste} \\ \rightarrow \text{protects itself due to alarm} \end{cases}$

eg: crocodile and bird
 $\begin{cases} \rightarrow \text{shining teeth} \\ \rightarrow \text{eat leftover food} \end{cases}$

eg: Hermit crab and sea anemone

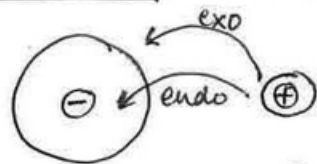
shelter — 
 protection — 

$\begin{cases} \rightarrow \text{(Proto-cooperation)} \\ \rightarrow \text{commensalism} \end{cases}$

Negative / Antagonism / Detrimental

exploitation (+/-)
 $\begin{cases} \rightarrow \text{Parasitism} \\ \rightarrow \text{Predation} \end{cases}$

I. Parasitism: (+/-)



Host

Parasite

+ growth
 + survival
 + reproduction

↓

Evolution

→ Parasitic adaptation

→ absence of unnecessary sense organ

→ Presence of adhesive organ and sucker

→ simplified digestive system (N loss of digestive system)

* → High & reproductive rate

(but have complex lifecycle even though morphologically/anatomically simpler)

>> Parasite can use one, two or more intermediate

i) human liver fluke

- snail
- fish

ii) Malarial Parasite — ♀ mosquito because it does not take its nutrition and shelter ~~on~~ from/on our body. They take our blood, to incubate their eggs and to keep their body warm.

characteristics of good parasite:

→ Parasite donot directly kill their host and depends on host for food and shelter

→ They are helpful for predators by making its host weak

Yadpa Yadpa ke maarna

Types

Ectoparasite

Ectoparasite

eg: → Human lice

Copepods on marine fishes (N)

→ sand fly

Ectophytoparasite

→ Red cotton bug

→ Plant lice (Aphids)

→ lac insects

Endoparasite

→ Tapeworm

→ Liverfluke

→ Taenia

→ ascariis

→ Entamoeba

→ Plasmodium (RBC)

* Endoparasite are more common than ectoparasite

Special

Hyperparasite: Parasite on Parasite

eg: Bacteriophage on Bacteria

Holoparasite:

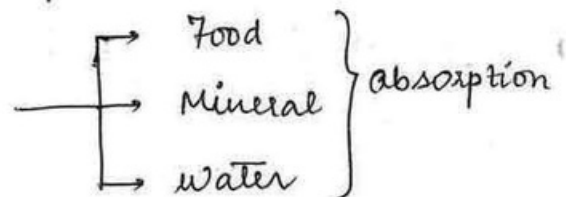
(complete parasite)

eg i)



Cuscuta: total stem parasite

sucking roots
or
Haustorial roots



ii) Rafflesia: total root parasite



Rafflesia

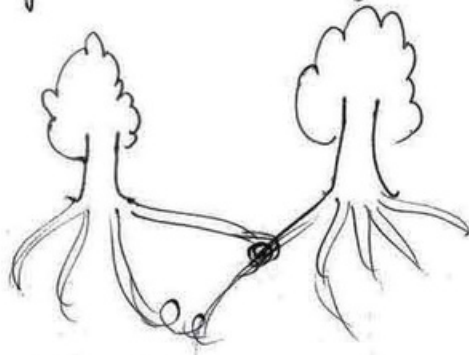
Hemiparasite :



food formation by itself
but depends on host for
mineral and water

eg: Viscum : oak
Loranthus : mango } Partial stem parasite

eg: Santalum } Partial root parasite



Brood parasitism :

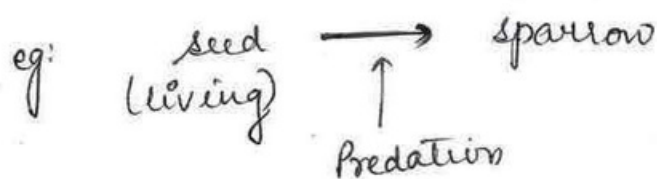
eg: Cuckoo and crow $\Rightarrow$ social Parasitism

★ ★ Not a true parasite

~~Indo~~ **II. Predation:** ★★ ★★

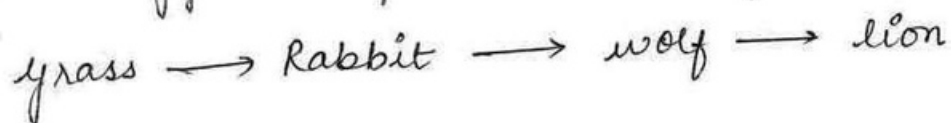
catch $\rightarrow$ kill $\rightarrow$ eat (hunting of living)

eg Rabbit $\rightarrow$ lion
 (small) (large)
 $\downarrow$ $\downarrow$
 Prey Predator



Significance :

- > Energy flow: Predator acts as a conduit (source) for energy transfer across trophic level.



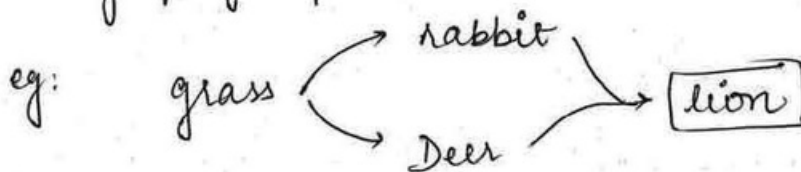
- > Maintain ecological balance: by controlling the prey population.

eg: 1920s - Australian :: (exotic)
 Prickly Pear cactus $\Rightarrow$ Natural Predator was ~~Out~~
 causes havoc by spreading millions of hectare
 (huge disturbance) rangeland.

$\Rightarrow$ Predator: Moth which brought it under control

- > maintains species diversity: by reducing competition among prey species

inter intra



eg: American pacific coast $\Rightarrow$ ~~(exotic)~~ Pisaster fish (Predator)
 (enclosed area)

when predator was removed, 10 species of invertebrates extinct within a year due to

> Biological control method

To control a living from other living for human welfare

→ mosquitoes controlled by Gambusia fish
(eat mosquito larva)

→ Birds controlled by hawk

→ Insects controlled by Dactylia and Anthro-
botrys (fungus) → Ectomorphous

∴ Predator is of PRUDENT nature

⇒ defense against Predation : (70% Animal) (20% Plants)
⊕_{nt} in biodiversity

Animals:

i) frog > Camouflaged (just the colour change)
Insects

ii) Monarch ~~bat~~ butterfly:

⊕_{nt} of special chemical, ∴ highly distasteful for its predator (bird) because at the caterpillar stages the butterfly eats poisonous weed (itself it was adapted) i.e. MIMICRY → change in shape

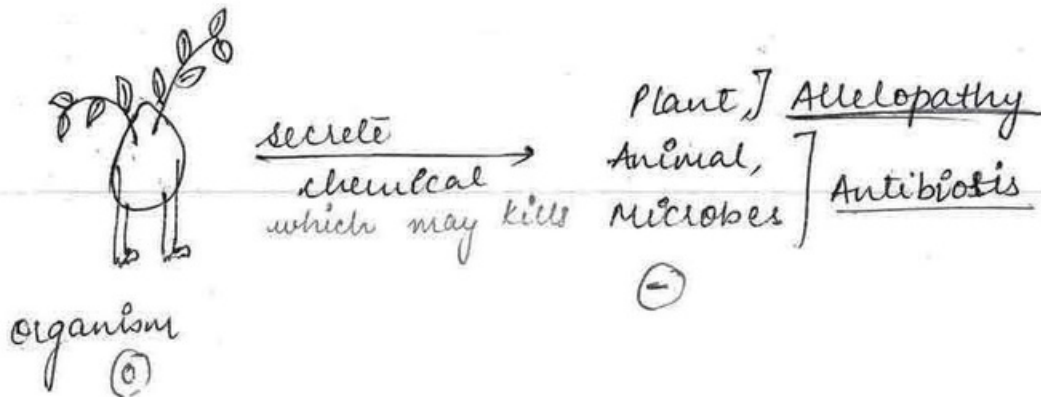
Plants:

i) Acacia > thorns
cactus

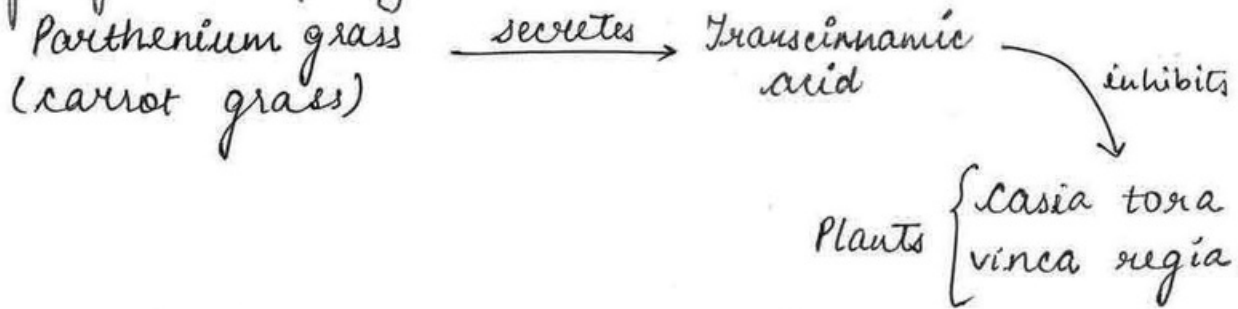
(N) ii) carotenis — cardiac glycosides chemical

iii) Nicotine, caffeine, Quinine, strychnine, opium.] Product
(Tobacco) (coffee) (cinchona) (Nuxvomica) (Poppy)] Plant

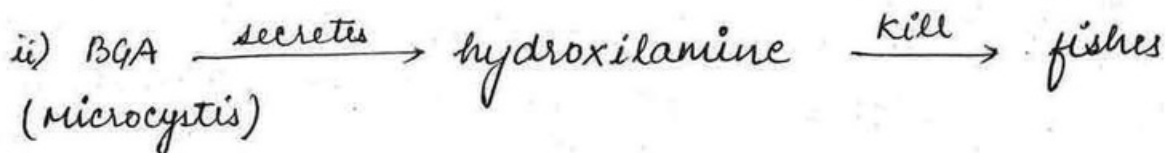
AMENSALISM : (0/-)



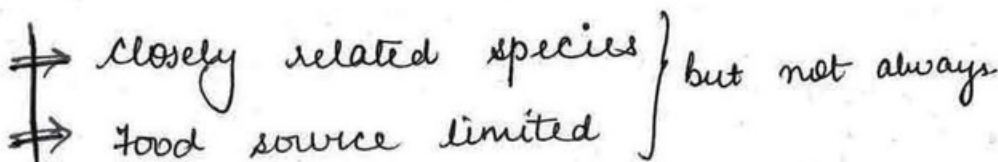
eg: of Allelopathy:



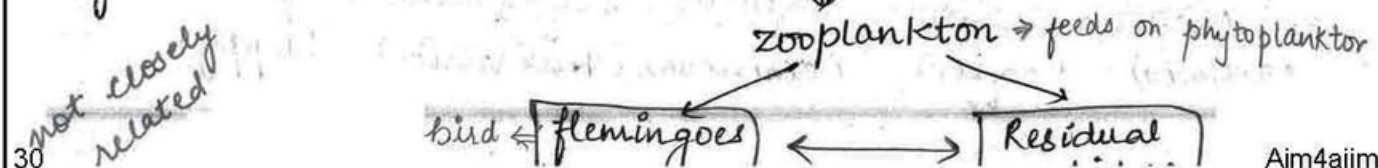
eg of Antibiosis:



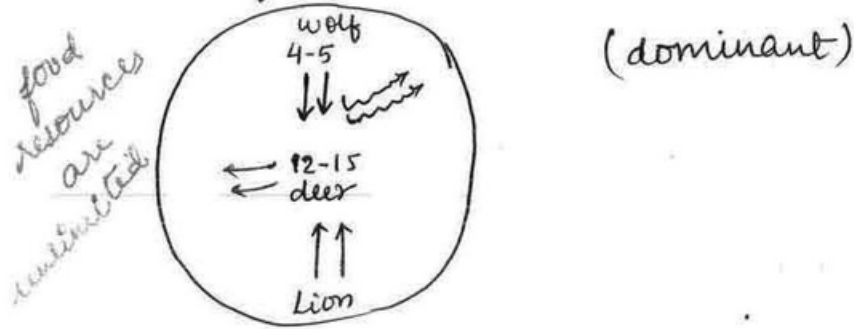
COMPETITION : (-/-)



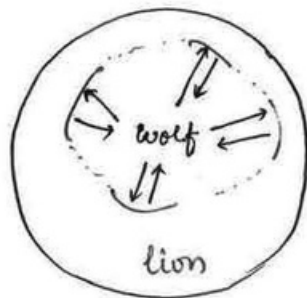
eg: south america: shallow water lake



i) Interference competition (indirect competition)



ii) competitive release:



Dominancy due to ⊕nce of superior species

iii) Gause's competitive Exclusion Principle.

↓
food source limited

↓
niche overlapping absent

superior

↓
inferior

remove

Conell's

exp. on scotland

↓
Rocky intertidal area

↓
Two barnacles species

superior: Belanus

inferior: Chathamalus

∴ when food sources are unlimited then niche overlaps.

extinct

eg: galapagos island

↓
Albington tortoise

superior grazer, → goat introduced

∴ All the tortoise extinct in a decade due to food scarcity

iv) Resource Partitioning: (Resource distribution)
food sources unlimited



Niche overlapping ⊕nt



⇒ feeding time and ^{eating pattern} foraging pattern changes

Mac Arthur → single tree
↓
5 birds of
(Warblers)

* In the ⊕nce of superior species, n ↓ of inferior species (i.e growth rate ↓es)

Ecosystem term : Tansley

(Biotic + abiotic)

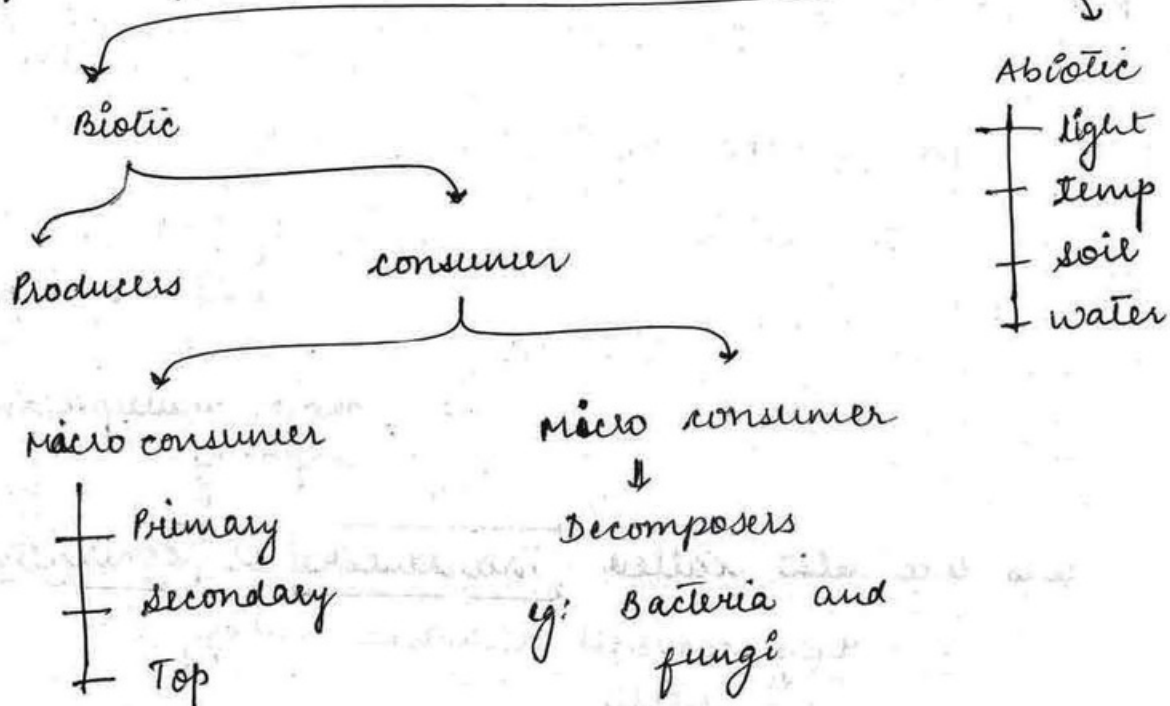
- Small and large
- Natural and artificial
- complete and incomplete
- Permanent and Temporary

Natural : Permanent/complete → self regulated
self maintained

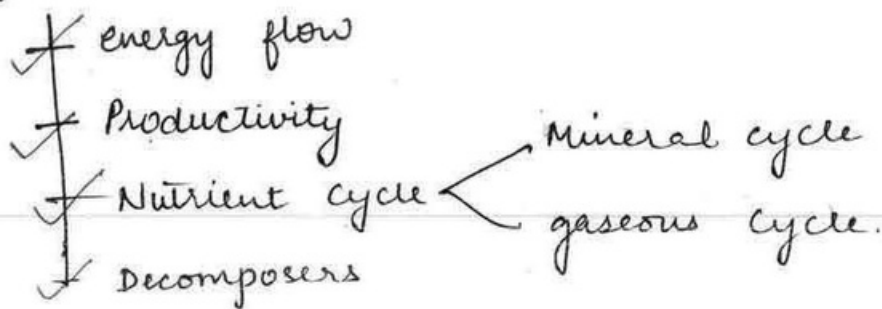
Artificial / man made : Temporary / Incomplete
eg: aquarium

* small → nanoecosystem → drop of water
large → megacosystem → eg: ocean

Acc to Oden: smallest structural and
father of ecosystem function unit



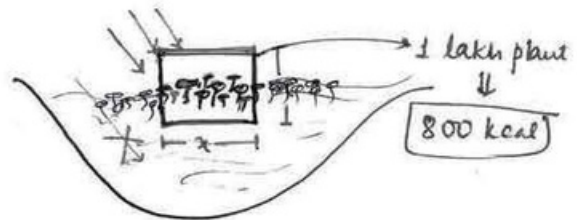
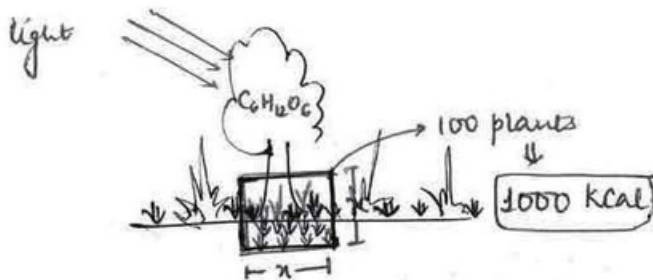
functional unit



Producers : Autotrophic / green plants:

Macrophyte
(Terrestrial)

microphyte
Phytoplankton (aquatic).



* Food production (UNIT AREA in UNIT TIME)
i.e. Productivity → **Terrestrial > Aquatic** per unit area and time

whereas for overall comparison,

Terrestrial < Aquatic $\frac{2}{3}$ of the earth's atmosphere and more multiplication capacity

* Producers are also called **Transducers** or **converters**
∴ they convert chemical energy into solar energy

CONSUMERS

Macroconsumer

Primary consumer
↓
directly depends on plants

Herbivores

eg: Cow, Rabbit, Deer, Rat
↓
seed

secondary consumer
↓
depends on primary consumer

Carnivores

Predator (livings) eg: Dog, cat, snake, wolf, fox etc
Scavenger (dead)

* Note: whale → herbivorous by nature
Phytoplankton + Zooplankton

Top consumer
↓

They can eat others but not eaten by others

Carnivores Omnivores

Plants
Animals
Man and Peacock

eg: lion, man, shark, Hawk, eagle, Peacock

* All predators are carnivores (X) ∵ predators can be herbivorous as well eg: Bird eating seed.

* All carnivores are predator (X) ∵ some of the carnivorous are scavengers as well.

⇒ Animal parasites are secondary consumer eg: sandfly, leech
⇒ Plant parasites are primary consumer eg: cuscuta, Rafflesia

⇒ Insectivorous plant

Photosynthesis (Producers)

Insect eating ⇒ (N₂ deficiency)
(sec. consumer)

⇒ Lion : Always top consumers

* ⇒ Human :
 → Primary consumer
 → Secondary consumer
 → Top consumer

i) Plant → man

ii) Plant → Goat → Man (secondary)

iii) Plant → Rabbit → wild pig → man / Top

⇒ sparrow → (Pri.) → seed
 → (Sec.) → insects } omni.

⇒ crow → (Pri.) → Bread
 → (Sec.) → Rat } omni.

⇒ cockroach → (Pri.) cellulose
 → (Sec.) insects } omni

⇒ ^{Organism using} Milk product → secondary consumer

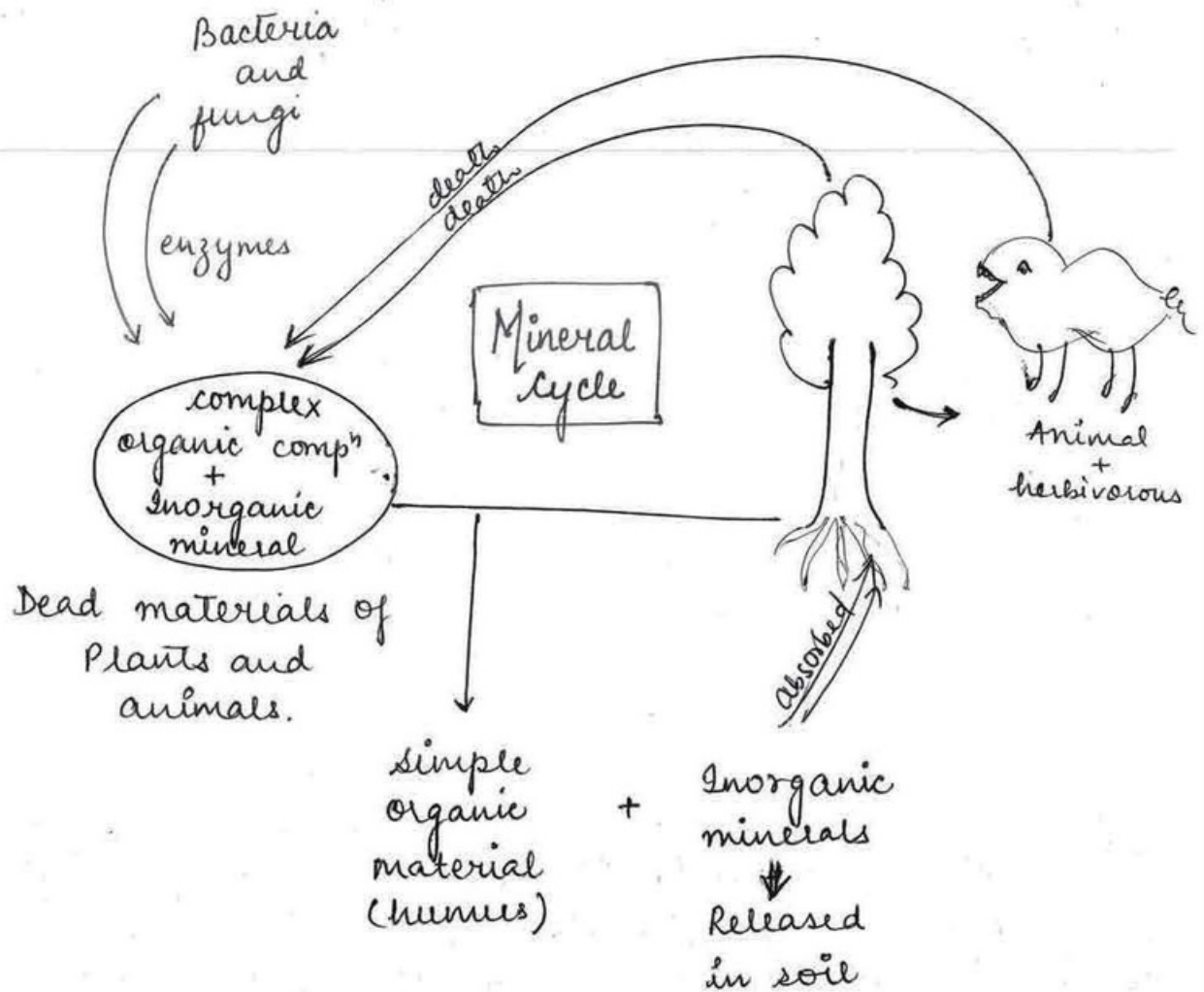
* ⇒ ~~Self~~ → sec. consumer

grass → cow → calf depends on cow for milk
 Microconsumers

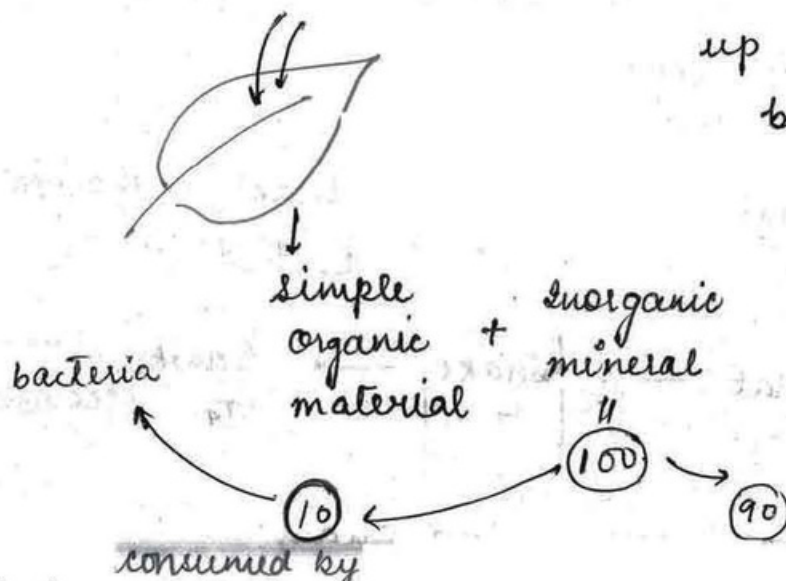
Decomposers → bacteria & fungi
 first digestion then ingestion ; Osmotroph / Saprotroph
 Transformer / Reducers : convert complex ~~sub~~ substances into simple

⇒ ∴ gets energy from oxidation, hence reduces others i.e. reducers

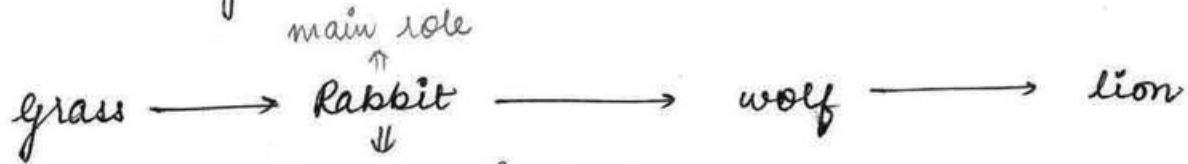
Karl Mobius : Biocoenosis (= ecosystem / community)



Nutrient immobilization : out of total minerals, some minerals are tied up with microbial biomass and that are separated from mineral cycle, called nutrient immobilization

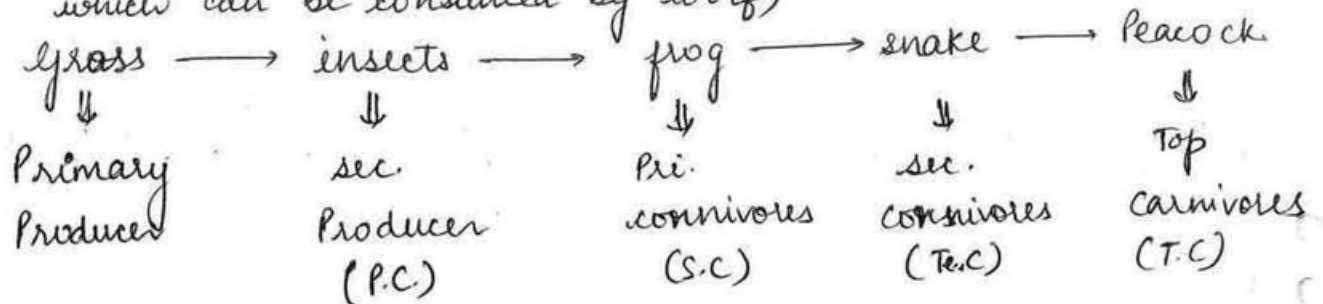


Secondary Producer:



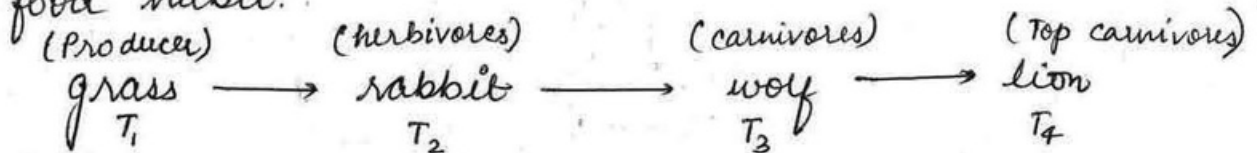
Secondary Producer
(Key Industry animal)

(it converts cellulose into proteins or carbohydrates which can be consumed by wolf)



Food chain:

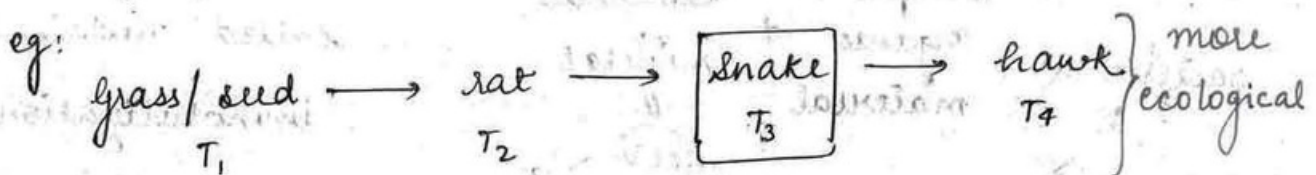
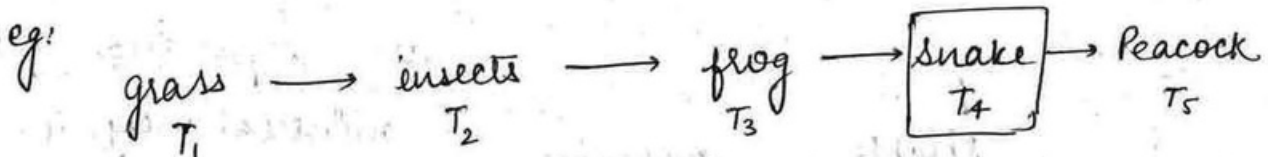
Sequence of organism according to their food habit.



Trophic level:

specific place of an organism in food chain

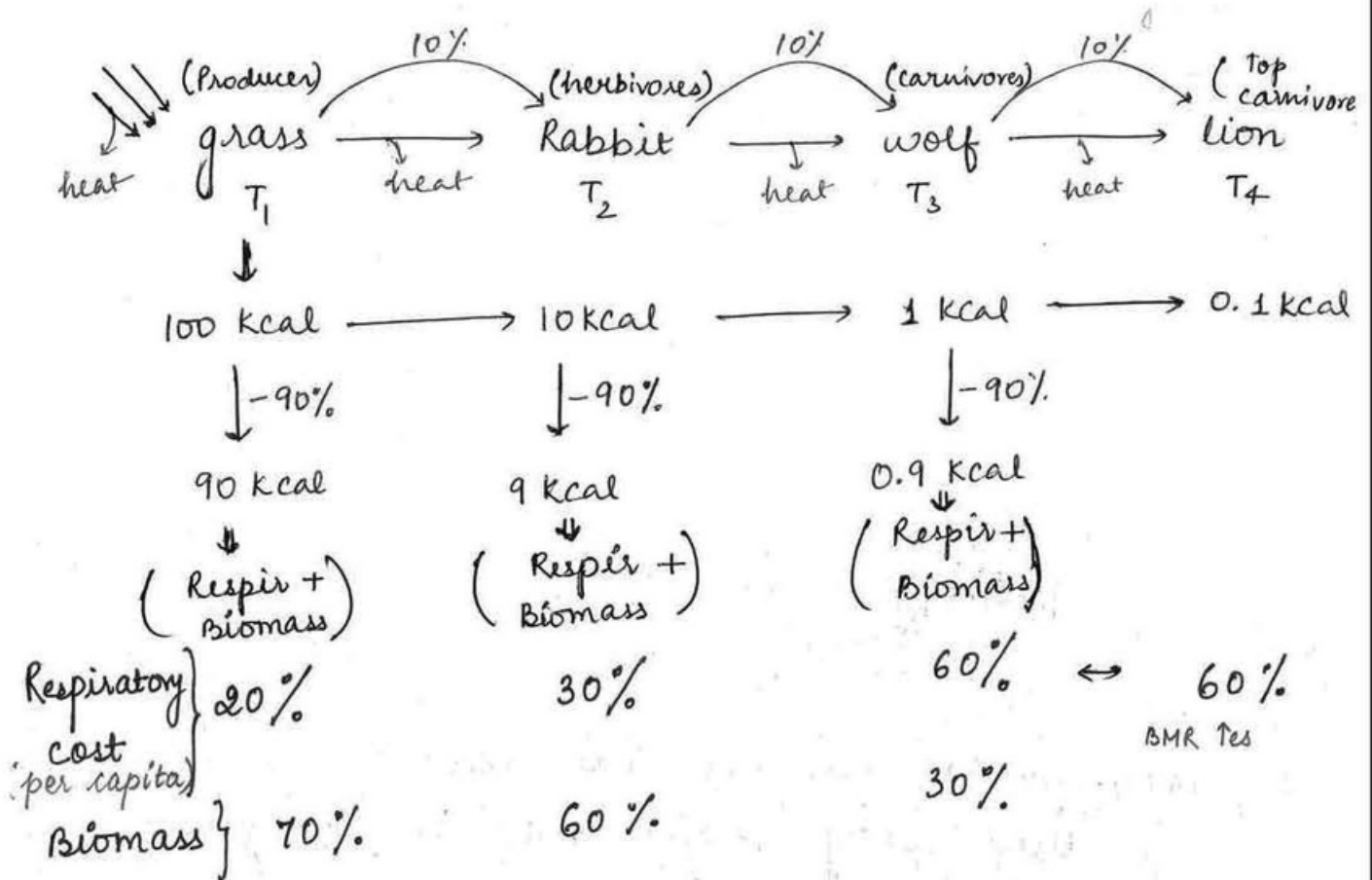
~~Trophic~~ Trophic level represents functional level
not a species level



* 3-4 trophic level → healthy food chain

because, energy transferred in Trophic level is never 100% efficient. It follows 10% energy transfer law of Lindeman

* growth + reproduction : Biomass
(1 rabbit) (group of rabbit)



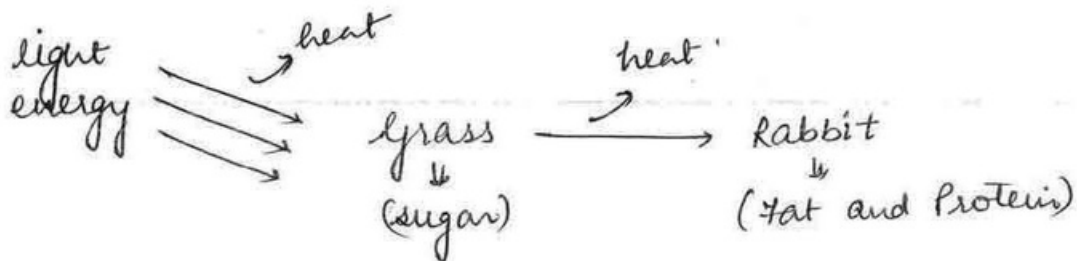
In successive trophic level :

energy → ↓↓es [100 → 10 → 1 → 0.1]

respiration cost ↑↑es [20% → 30% → 60%] to maintain internal energy

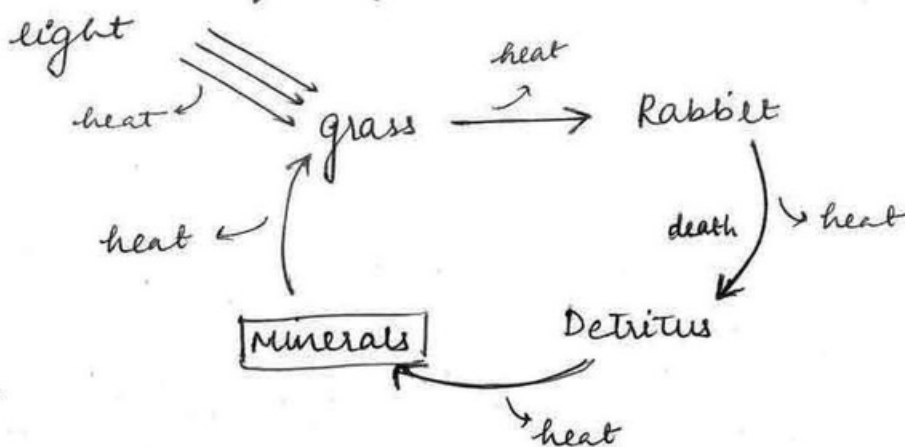
- ⇒ Top consumers are ecological weakest but physically strong. cz it is connected to many food chain. (part of food web)
- ⇒ Ecological strong → Producer (has more amount of energy)

1st law:

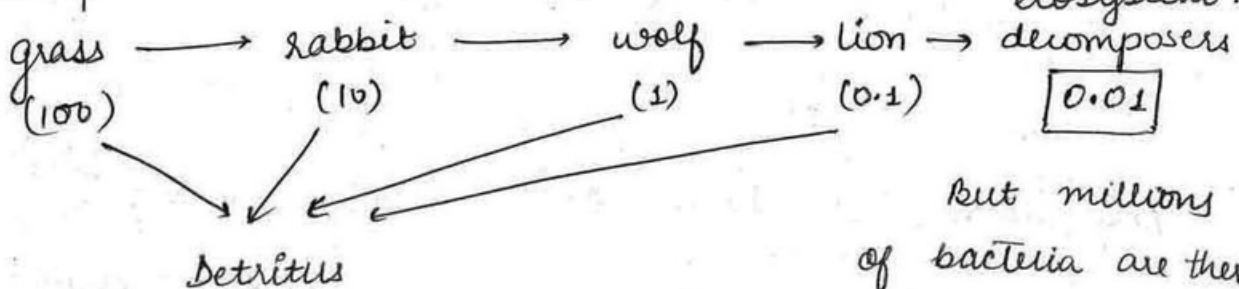


2nd law: heat loss.

Transfer of energy is always UNIDIRECTIONAL but transfer of minerals is in cyclic manner.



* Decomposers are not the part of food chain but they play major role in ecosystem. i.e. producers and decomposers are essential in an ecosystem.



But millions of bacteria are there so negligible energy reaches upto bacteria so these are not considered in food chain.

Types of food chain :

- ① grazing food chain / Predatory food chain :
small $\longrightarrow$ large (size of organism)

terrestrial grass $\longrightarrow$ rabbit $\longrightarrow$ wolf $\longrightarrow$ lion

aquatic

Phytoplankton $\longrightarrow$ zooplankton $\longrightarrow$ small fish $\longrightarrow$ large fish (shark)

- ② Parasitic food chain :
large $\longrightarrow$ small

Tree $\longrightarrow$ birds $\longrightarrow$ Parasite [top consumer]

- ③ Detritus food chain : (exception) $\because$ doesn't ~~follow~~ follow 10% loss. and unidirectional flow.

Terrestrial

Detritus $\longrightarrow$ bacteria and fungus
(Plant + animal dead material)

Aquatic

Detritus $\longrightarrow$ bacteria and fungi $\longrightarrow$ ~~leaves~~ larva of mollusc and nematods

large fish $\longleftarrow$ small fishes
bird $\longleftarrow$

* Sunderbans :
~~doesn't follow~~
10%.

Mangrove trees $\longrightarrow$ Detritus $\longrightarrow$ Bacteria and fungus
Tiger $\longleftarrow$ small fish $\longleftarrow$ larva of mollusc

~~Somehow~~

Detritus food chain can form a link with grazing food chain.

eg: Detritus → Detritivores → bird → Hawk
eg: earthworms,
ants and
Termites

eg: Detritus → Bacteria and fungus → larva of mollusc and nematods → small fish → large fish

⇒ In terrestrial condⁿ: Max energy flow in
DETRITOUS FOOD CHAIN
(∵ ^(detritus) nonliving > living) on land

⇒ In aquatic condⁿ: Max energy flow in
GRAZING FOOD CHAIN
and grazing of zooplankton
of phytoplankton is very high
↳ multiplication
↳ less decomposers

* Producer should have higher biomass so that the food chain may survive.

* Long term stability of ecosystem depends upto predation

* At any trophic level, no. of organism depends on, no of organism in previous trophic level (i.e. availability of food)

⇒ food chain do not exist ^{naturally} whereas food web does

DFC

→ Magnitude (distribution of energy) → high

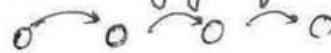


* slow energy flow

GFC

Magnitude (low)

fast energy ~~flow~~ flow

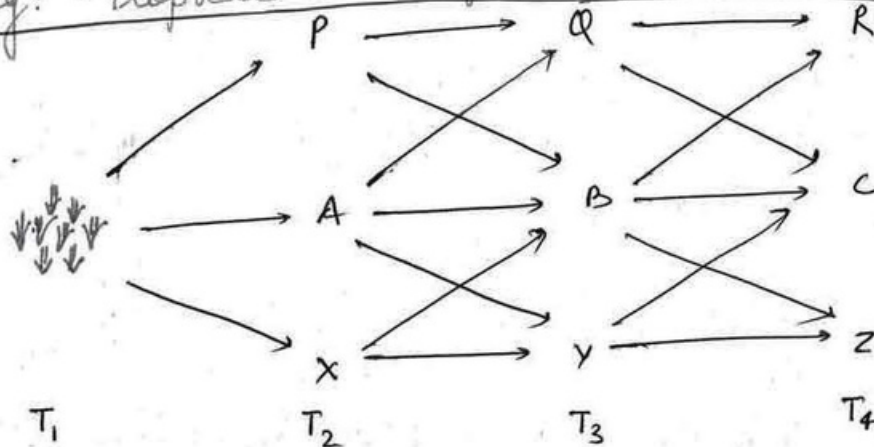


Food web:

Network of interconnected food chain in biotic community. [facility of taste and choice]

* Organism does not feed on similar trophic level

* as no of ~~organisms~~ ^{species} ↑ ∴ diversity ↑
food web complexity & stability ↑
eg: Tropical rain forest



* as no of ~~organisms~~ species ↓ ∴ diversity ↓
i.e. simple food web is less stable/unstable
eg: Arctic

generally,

✓ Natural ecosystem $\Rightarrow$ complex food web

✓ Artificial ecosystem $\Rightarrow$ Food chain
eg: aquarium, cropland.

*Q) Ecosystem creates:

i) food chain

ii) food web

iii) Both

iv) Any of 2

i) food chain

OR ~~ii)~~ food web (priority)

iii) Both

iv) none

$\therefore$ type of ecosystem

i.e. (natural or artificial)

is not mentioned $\therefore$

we can consider both
but one at a time

Both option is not correct

$\therefore$ at a time we can
consider one ecosystem
i.e. either ~~natural~~ or
artificial

Q) Ecosystems create:

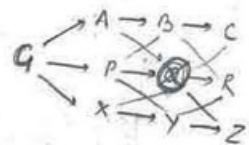
i) food chain

ii) food web

iii) Both i.e. artificial as well as natural.

iv) none

Homeostasis: self control / self balance



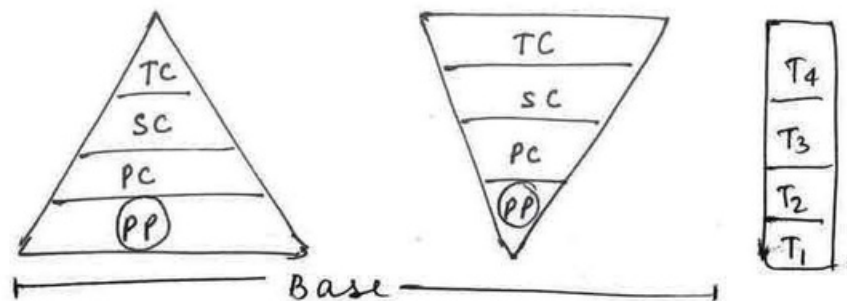
* Science that deals with self regulation of
ecosystem is called CYBERNETICS

(maintains equilibrium)

Ecological Pyramids:

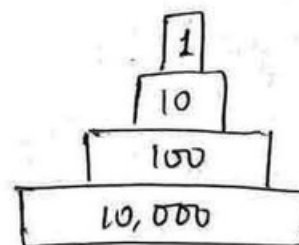
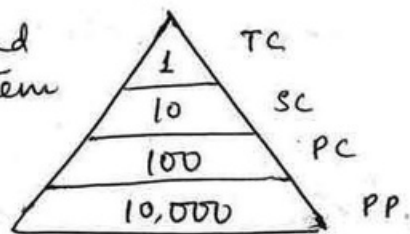
Graphical representation of ecological parameters like number, energy and biomass.

	grass	→	Rabbit	→	wolf	→	lion
number:	10,000		100		10		1
Energy:	100 kcal		10		1		0.1
Biomass:	120 kg		80 kg		50 kg		20 kg



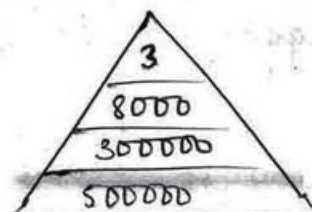
i) Pyramid of number:

grassland ecosystem



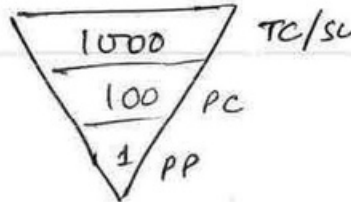
aquatic ecosystem

Phytoplankton → zooplankton → SF → LF



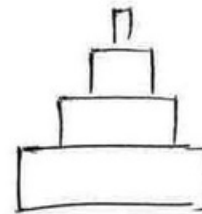
tree ecosystem]

Tree $\longrightarrow$ Birds $\longrightarrow$ Parasites
1 100 1000



Note: forest ecosystem]

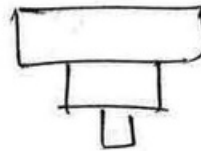
Producer $\longrightarrow$ PC $\longrightarrow$ SC $\longrightarrow$ TC
(Tree shrubs herbs etc)



forest ecosystem] $\Rightarrow$ Parasitic food chain

Tree dominating

Tree $\longrightarrow$ bird $\longrightarrow$ Parasite



i) upright pyramid of number absent in:

i) lake $\triangle$

ii) Pond $\triangle$

iii) Forest $\triangle$ / ∇ ✓

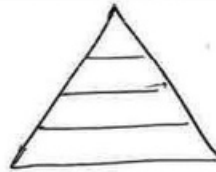
iv) grass $\triangle$

ii) Pyramid of energy
always upright.

represents the productivity of ecosystem as well as

iii) Pyramid of ^{weight} biomass :

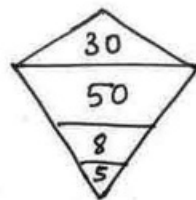
grassland] grass $\rightarrow$ rabbit $\rightarrow$ wolf $\rightarrow$ ~~lion~~ lion
 120 kg 80 kg 50 kg 20 kg



* In grassland ecosystem, ~~the~~ all the pyramids (i.e. energy, number and biomass) are always UPRIGHT

AIMS 2005 Aquatic] Phyto $\rightarrow$ zooplank $\rightarrow$ SF $\rightarrow$ LF
 5 kg 8 kg $100 \times \frac{1}{2}$ kg 6×5 kg
 (50 kg) (30 kg)

* Unit area
Unit time

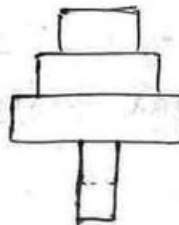
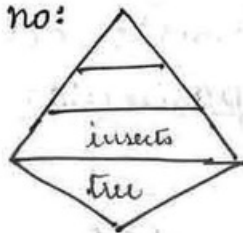


spindle shaped

* Hypothetical Pyramid:

Tree $\rightarrow$ insects ~~insect~~ $\rightarrow$ insectivorous birds $\rightarrow$ large bird (hawk)

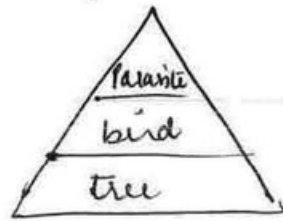
Pyramid of no:



* When unit time and unit area are not considered, then the pyramid MAY be upright

tree ecosystem]
Biomass :

Tree $\longrightarrow$ Birds $\longrightarrow$ Parasite



* Most of the pyramid are upright

$\Rightarrow$ Inverted

Number $\Rightarrow$ tree ecosystem

Biomass $\Rightarrow$ Aquatic ecosystem

* Pyramid of no represents biotic potential

* Pyramid of biomass represent standing crop

$\Downarrow$
(total living organic material)

$\Rightarrow$ standing crop:

Total amount of living organic material $\oplus$ nt in particular area in particular time is standing crop. It is also considered at a trophic level basis

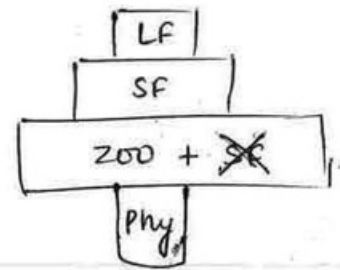
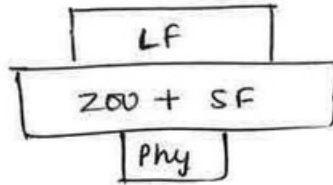
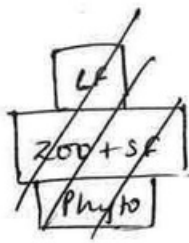
$\Rightarrow$ standing state:

Total amount of inorganic material (non-living) $\oplus$ nt in particular area at particular time

limitations:

i) Only ~~at~~ food chain are considered, i.e. food webs are not included in pyramid

ii) Decomposers (detritus food chain) are not considered



iii) Organism that comes more than one trophic level at a time are not considered in pyramid.

eg: Insectivores plant (Primary producer + Sec. consumer,

Productivity

Rate of Production.

weight	energy:
$gm\ m^{-2}\ yr^{-1}$	$Kcal\ m^{-2}\ yr^{-1}$

Primary (Producer) $\Rightarrow$ amount of biomass or organic matter produced per unit area over a time

Secondary (consumer)

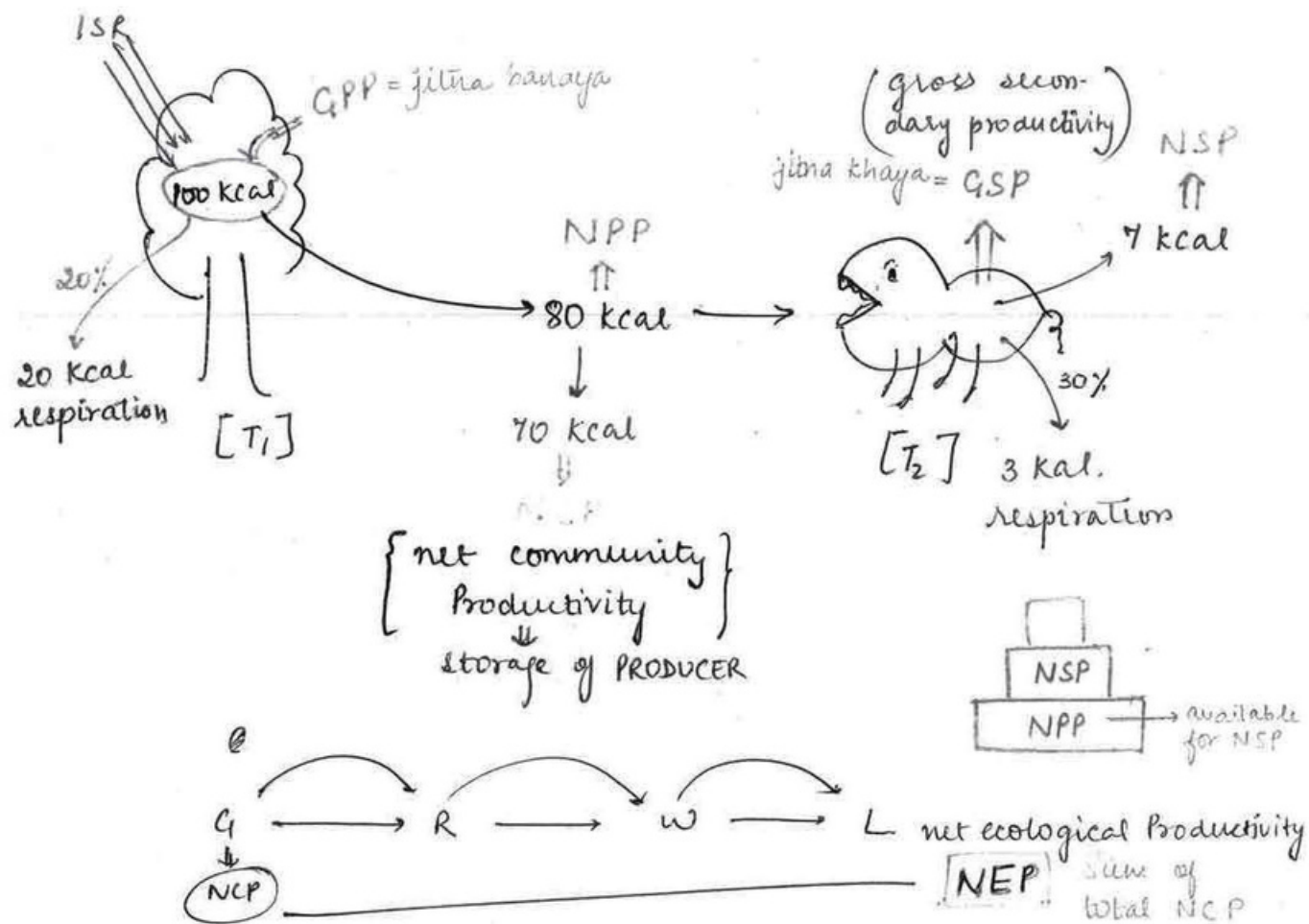
Primary consumer produces food for secondary consumer

GPP
gross
primary
productivity

NPP
Net
primary
productivity

$\Downarrow$
Total food for^m by producer

$\Downarrow$
Energy left after respiratory loss



GPP: Total amount of organic food form^m by producers & in unit area and unit time
 → Total photosynthesis / Total fixation

NPP: Total amount of stored food in producer after respiratory loss.
 → Rate of new tissue formation

NCP: Total amount of stored food in producer after heterotrophic use / consumption
 consumer
 i.e. 10% loss

$\therefore \boxed{NPP = GPP - R}$ $R = \text{respiration} + \text{metabolic activity.}$

$\boxed{NCP = NPP - HR}$ $HR = \text{used by heterotrophs}$

Factors affecting productivity:

① Plant inhabiting in a particular area

Terrestrial > aquatic

+ Tropical rain forest ↑
+ Desert ↓

+ coral reef ↑
+ deep oceans ↓

② environmental factor:
Most effective $\Rightarrow$ light (more photosynthesis)

③ Minerals $\begin{cases} \rightarrow \text{ocean} \Rightarrow \text{N}_2 \text{ deficiency} \\ \rightarrow \text{lake} \Rightarrow \text{P deficiency} \end{cases}$ } diversity ↓

④ Photosynthetic capacity (unit area)
Terrestrial > aquatic

Maximum productivity $\Rightarrow$ Tropical rain forest

Tropical rain forest > coral reefs

↓
Deciduous forest

↓
coniferous

↓
Desert/Tundra

↓
lake

↓
river

↓
deep ocean

* Cropland:

sugarcane > Rice

↓
C₄ plant

$\therefore$ no photorespiration
i.e. no loss

Ecological efficiency:

$$T_1 \xrightarrow{\% = ?} T_2$$

$$\frac{T_2}{T_1} \times 100 \Rightarrow \boxed{\frac{GSP}{GPP} \times 100} \Rightarrow \frac{10}{100} \times 100 = 10\%$$

Net production efficiency: (at producer level)
(% of NPP)

$$\boxed{NPE = \frac{NPP}{GPP} \times 100}$$

Photosynthetic efficiency: amount of food form

$$\boxed{P_e = \frac{GPP}{ISR} \times 100}$$

ISR $\Rightarrow$ Incident solar radiation

① Annual rate primary productivity of biosphere $\Rightarrow$ 170 billion tons

② Productivity of ocean $\Rightarrow$ 55 billion tons almost 3 times

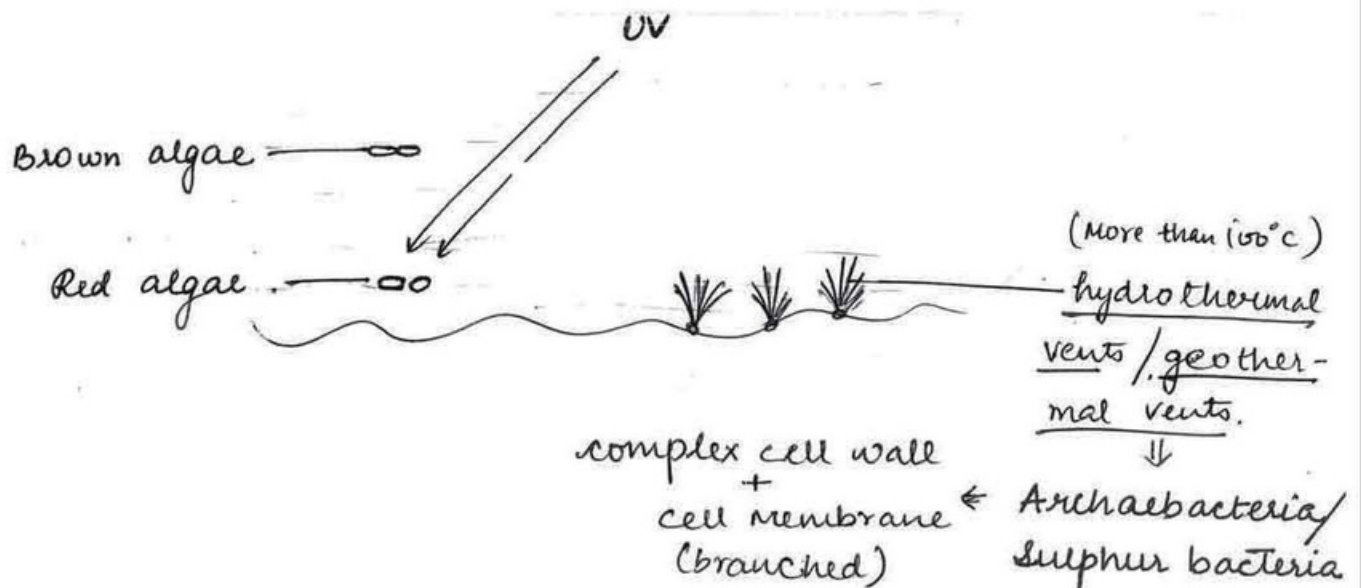
Productivity of land $\Rightarrow$ 115 billion tons almost 2 times

i.e. terrestrial > aquatic

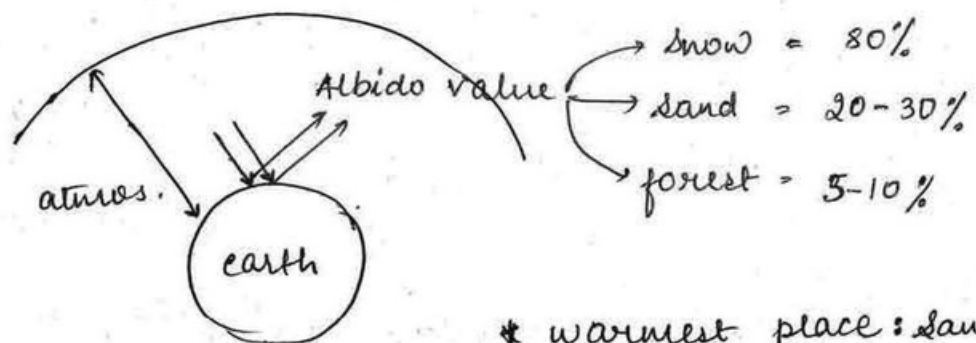
Abiotic :

1. light

Most effect — Productivity
Deep ocean — light $\ominus$



$2 \text{ cal cm}^{-2} \text{ min}^{-1}$ *



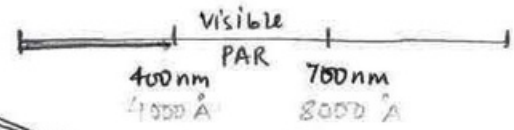
* warmest place: sand

* forest has lower Albedo than snow due to its

⊕ of its canopy, which absorbs max radiated light

ISR, K_i NPP

100 (10) 1-5%
50 (10) 0.8-4%
(20% used in respiration)



1-5% Photosynthesis

(less than 50%)

visible
(PAR)

Photosyn. active radiation

* (2-10%) photosynthesis

(more than 50%)

other

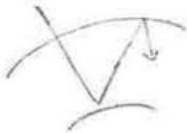
UV
Cosmic

X-Ray

γ -Ray

Infrared

$\lambda \downarrow$ energy $\uparrow$
(100-400 nm)



UV-C

(100-280) nm

most lethal
absorbed in
ozone formation
and destruction
 $\downarrow$

UV-B

(280-320) nm

skin cancer

mutation

Xeroderma
pigmentosum

snow blind-
ness cataract
(cornea
inflammation)

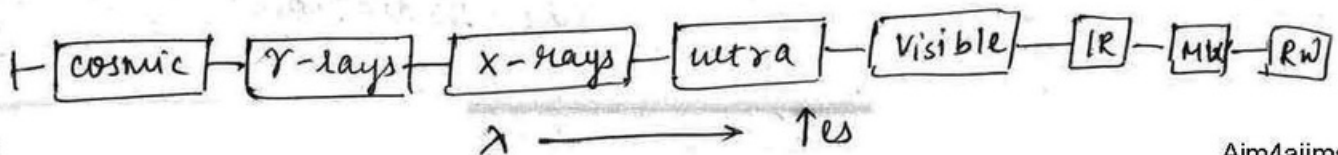
UV-A

(320-400) nm

light

Photokinesis: Regulation of speed of an organism
by the help of light called photokinesis

intensity $\uparrow \Rightarrow$ speed $\uparrow$



2. Temperature:

most effect $\rightarrow$ Diversity and distribution

i) Eff of Temp on plant:

$\Rightarrow$ Dessication: state of dehydration/Dryness

$\Rightarrow$ Chilling injury: burning of cell due to temp.
turns black $\rightarrow$ cells gets damage

high temp plant $\rightarrow$ low temp plant shock

$\Rightarrow$ freezing injury: low temp plant $\rightarrow$ Ultra low temp.

(Ice crystal form inside the plant)

ii) Eff of temp on morphology of animals

(m^2) surface area : vol^m (m^3)

Animal $\rightarrow$ ~~is~~ small size $\rightarrow$ vol $\downarrow$
S.A $\uparrow$ $\rightarrow$ heat loss $\uparrow$

(relative) large size $\rightarrow$ vol $\uparrow$
S.A $\downarrow$ $\rightarrow$ heat loss $\downarrow$

$\therefore$ Small animals are rarely found in cold area
Polar

Bergman rule:

Body size $\begin{cases} \rightarrow \text{cold area} - \text{Body size } \uparrow \\ \rightarrow \text{warm area} - \text{Body size } \downarrow \end{cases}$

Camel = Physiological adaption

Allen's rule: Cold area $\rightarrow$ eye, ear, snout, tail - small

warm area $\rightarrow$ eye, ear, snout, tail - large

iii) Effect of temp on Physiology:

Acclimatization: Physiological or metabolic adjustment of body according to changing environment is called acclimatization.

eg: Altitude sickness: Pressure ↓, oxygen ↓
symptoms: Nausea, Fatigue, heart palpitation

adjust → RBC count ↑
Hb - oxygen binding capacity ↓
Hb - oxygen ~~binding~~ ^{releasing} capacity ↑
Breathing rate ↑

eg: Antarctica fishes: Antifreezing Protein and glycerol
↓
body fluid (freezing point ↓)

cold area: Ice nucleating protein
↓

work as insulator ← Ice layer form in below skin

Aquatic mammals: Seal and whale
↓

work as insulator ← Blubber ← Thick fat layer

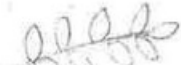
Kangaroo rat: Release concentrated urine to minimize water loss

- fat oxidation (H_2O)

do not drink H_2O

Plants: (Morphology adaptation)

a) Opuntia: leaves convert into spines

b) Prosopis etc:  Pinnate compⁿ leaves
- hypostomatic leaves
- sunken stomata (deep stomata)
- Thick cuticle layer

(Physiology adaptation)

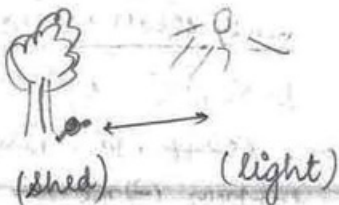
a) CAM pathway - scotoactive stomata $\left\{ \begin{array}{l} \text{Day - close} \\ \text{Night - open} \end{array} \right.$

b) Cells - Proline and sorbitol $\rightarrow$ cell fluid.
(AA) (Amino Acid)
 $\downarrow$
concentrated
 $\downarrow$
water loss $\leftarrow$ OP $\uparrow$ es

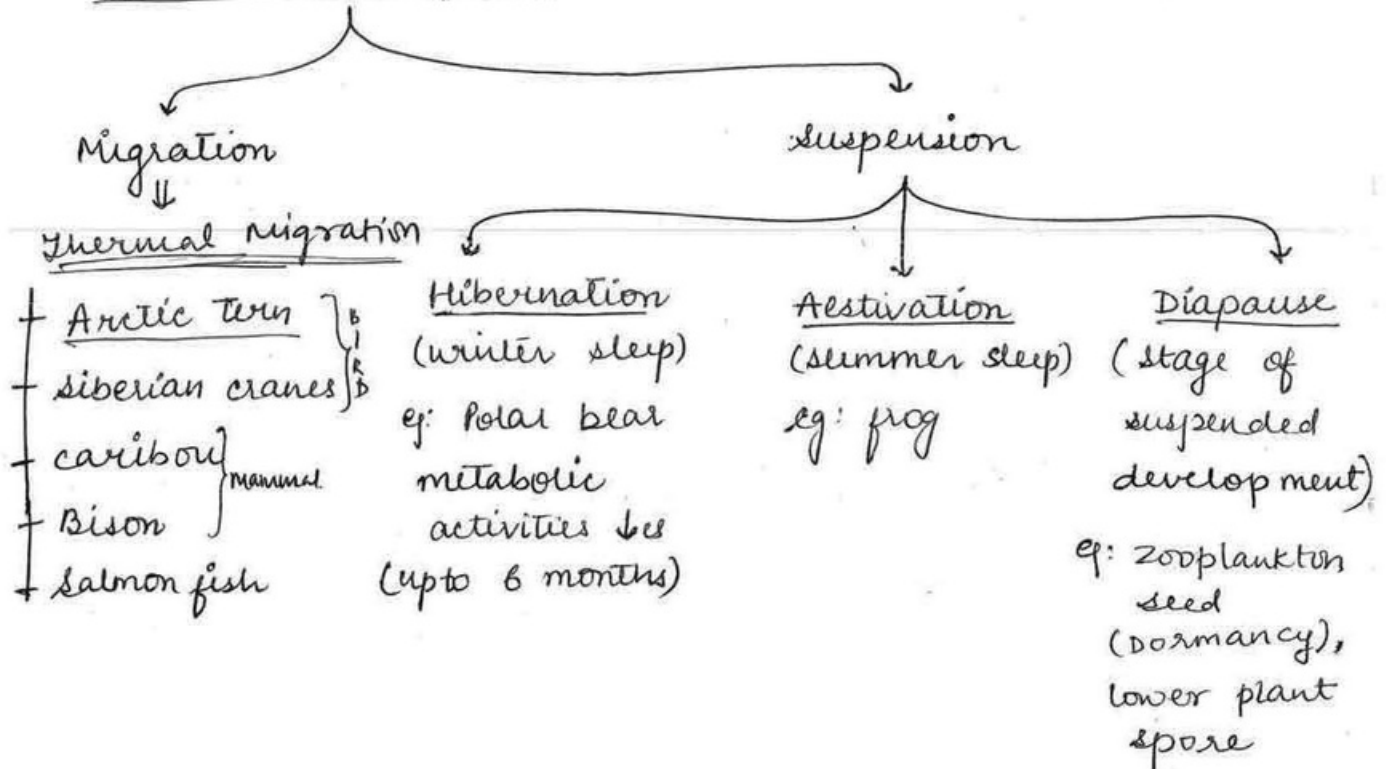
c) Chaperonine $\rightarrow$ Bind with protein $\rightarrow$ Prevent denaturation and maintain structural protein
(heat shock protein)

★ Behavioural adaptation:

eg) Desert lizard:
Physiological ~~behavi~~
adaptation $\ominus$ nt

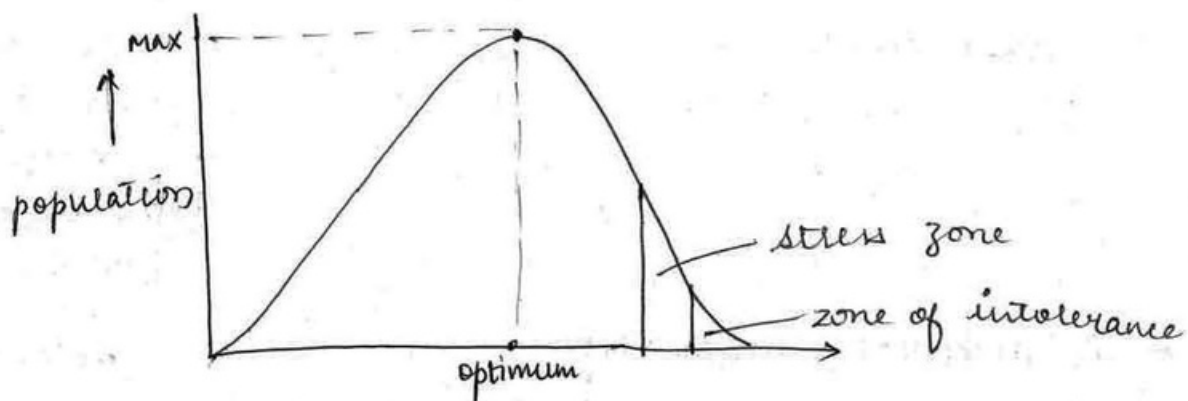


Behavioural adaption :



Shepherd's law of Tolerance

ecological amplitude : (Min-max) tolerance ~~rate~~ range



Types of ~~environment~~ ^{organism} on the basis of Temp :

EURYTHERMAL / warm blooded / homeothermal / endothermal
can tolerate wide range of Temp i.e. physiological adaptation ⊕
nt e.g. Aves, Mammals

STENOTHERMAL / cold blooded / poikilothermal / ectothermal
can not tolerate wide range of Temp. i.e. Physiological adaption ⊖
nt

stenothermal do not have large amount of energy to maintain its constant Temp.

The expenditure of energy in eurythermal is much higher to maintain constant ^{internal} temp.

On the basis of salinity:

Euryhaline
can tolerate wide range of salinity

e.g. Fishes of Estuaries
Osmoregulation

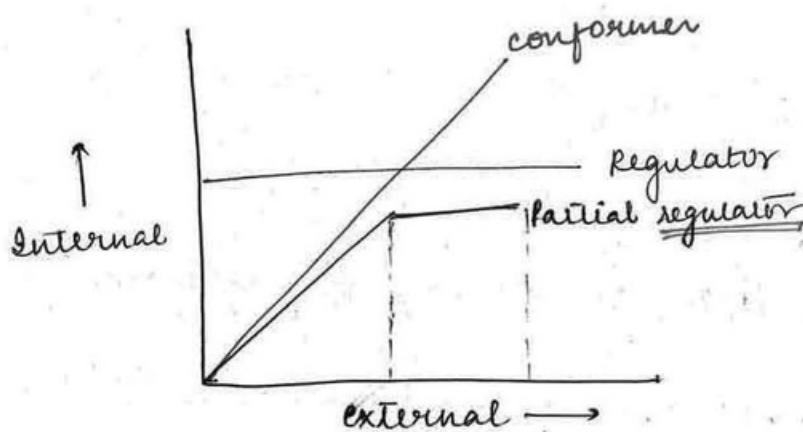
Stenohaline
cannot tolerate wide range of salinity.

① **REGULATOR**: Thermoregulator and Osmoregulator

e.g. Mammals, birds and fishes of Estuaries

② **CONFORMER**: Stenothermal and stenohaline

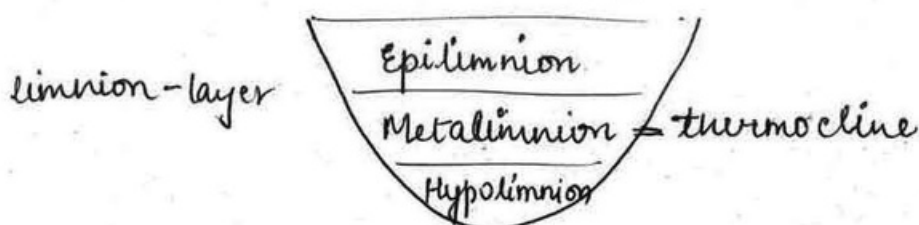
e.g. 99% animal and 100% plants.



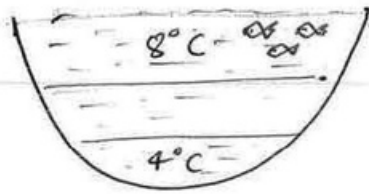
* [Brown Desert lizard]

≠ desert lizard

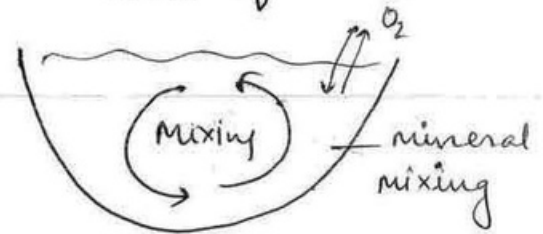
⇒ Thermal stratification in lake:



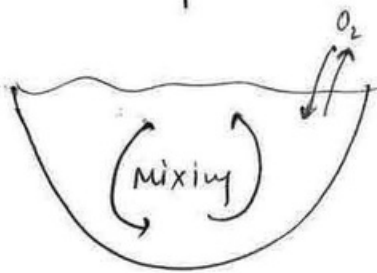
Summer stratification



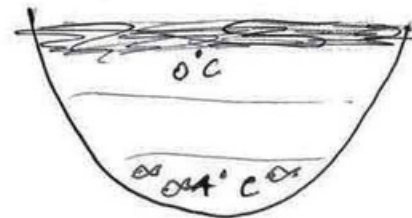
Autumn mixing ~~stratification~~



high O_2 and high mineral availability



Spring mixing

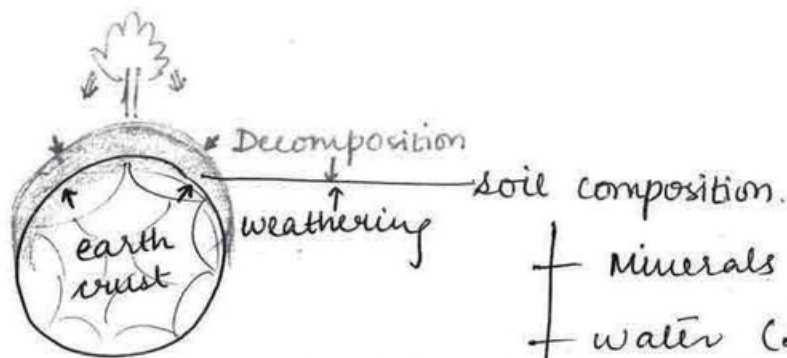


winter stratification

1 year $\rightarrow$ 2 stratification $\rightarrow$ summer, winter
 $\rightarrow$ 2 mixing $\rightarrow$ spring, Autumn
 (Max. growth of phytoplankton)

4°C $\Rightarrow$ water at max density ($\therefore$ always $\oplus$ at bottom)
 $\rightarrow$ not affected by external Temp difference

③ Soil



soil formation : Bidirectional

- + Minerals (45%)
- + water (25%)
- + air (25%)
- + organic material (5%)
dead + living

» soil particle size

clay ——— (less than 0.002 mm)

silt ——— (0.002 – 0.02 mm)

fine sand ——— (0.02 – 0.2 mm)

coarse sand ——— (0.2 – 2 mm)

gravel/grit ——— (2 – 5 mm)

coarse gravel — (more than 5 mm)

* loamy soil : 70% sand + 30% clay and silt

Best soil for
cultivation

{ 40% sand + 40% silt + 20% clay }

- + high water holding capacity (⊕nce of clay)
- + high air (⊕nce of sand)
- + high percolation

soil pH = (5.5 – 6.5)

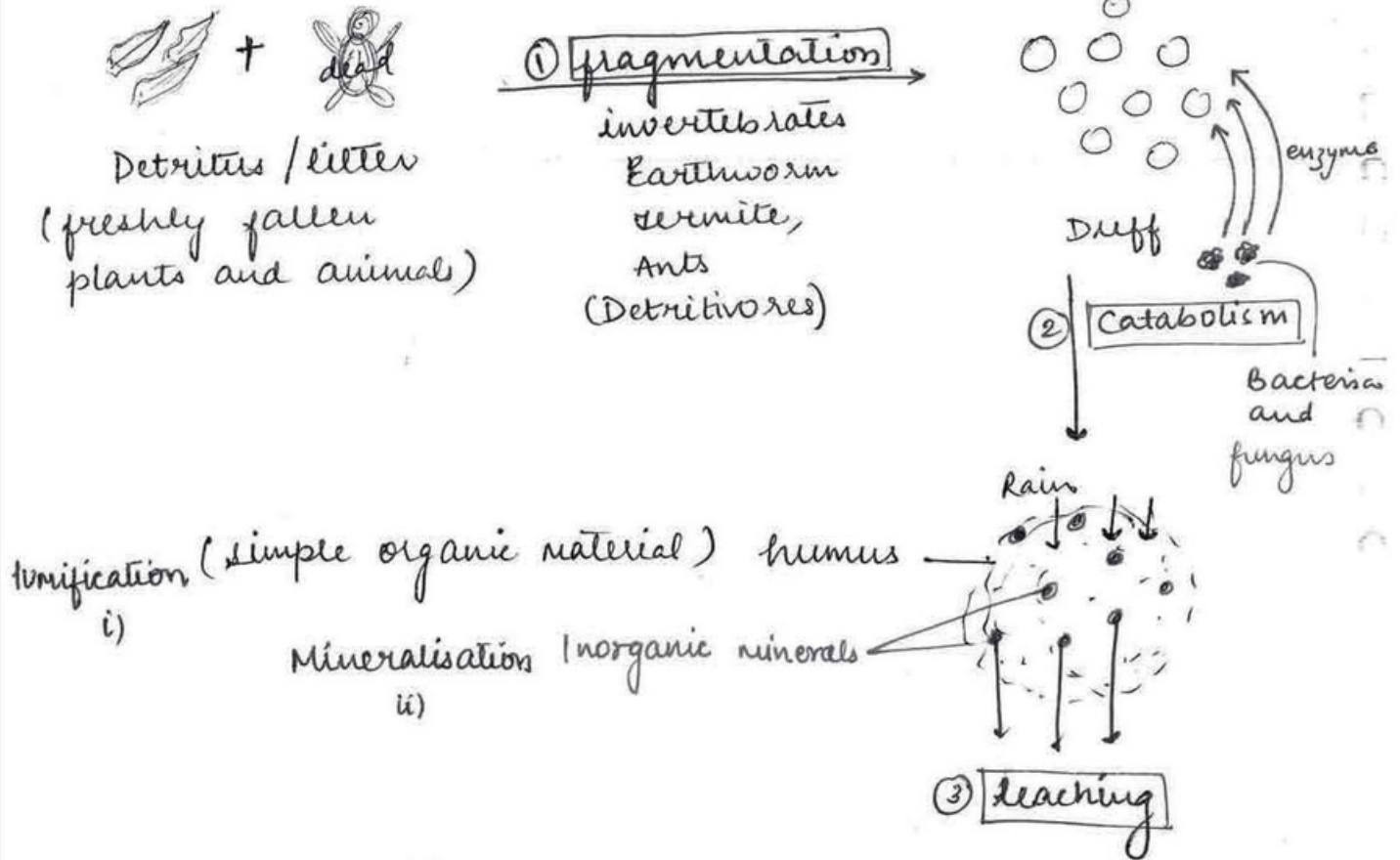
Decomposition :

Detritus → decomposition of plants and animal

Litter → " " plants only

Detritus → duff → humus

Decomposition



NCERT: ① fragmentation
② leaching
③ catabolism ← humification
mineralisation

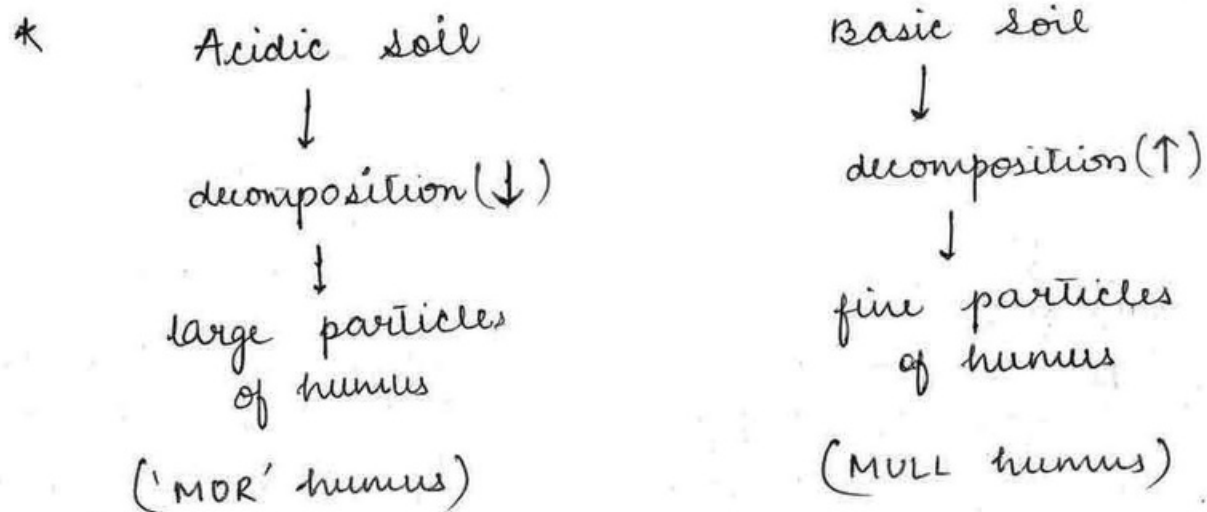
humus content gets resistant for bacteria (microbial action), Only mineralisation can take place

Factors:

① Temp: → temp ↑ ∴ Decomposition ↑
eg: Tropical rain forest
→ temp ↓ ∴ decomposition ↓
eg: Arctic

② Moisture: moisture ↑, decomposition ↑

- ③ lignin, chitin, suberine,
(complex substances) $\Rightarrow$ decomposition - slow
 - ④ Nitrogenous and sugar containing substances $\Rightarrow$
decomposition - fast
-
- ⑤ Aerobic $\Rightarrow$ decomposition - fast
 - ⑥ Anaerobic $\Rightarrow$ decomposition - slow
 - ⑦ Acidic soil $\Rightarrow$ decomposition - slow
(pH 3.8 - 4)
 - ⑧ Basic soil $\Rightarrow$ decomposition - fast



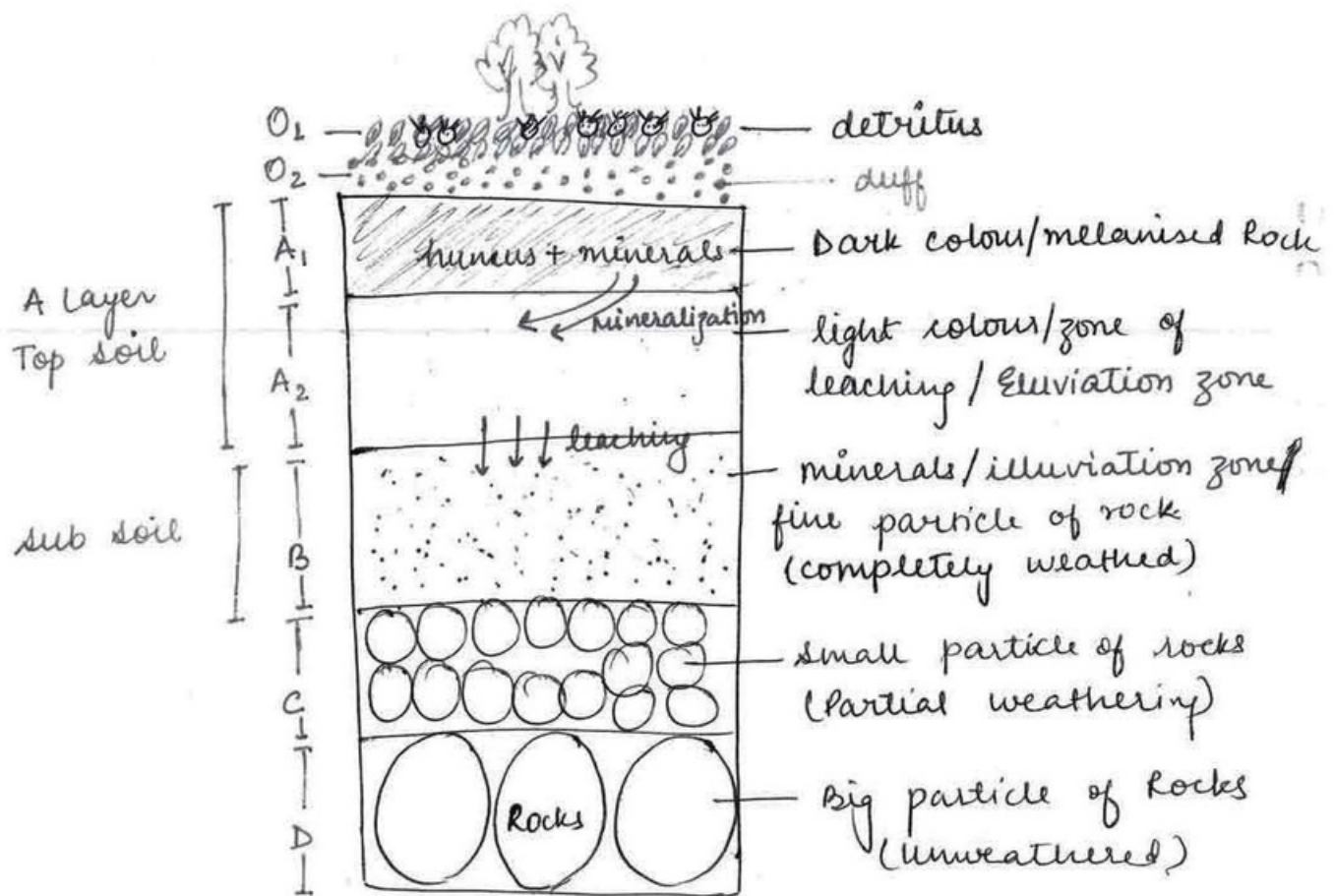
soil profile

Pedogenesis: formation of soil

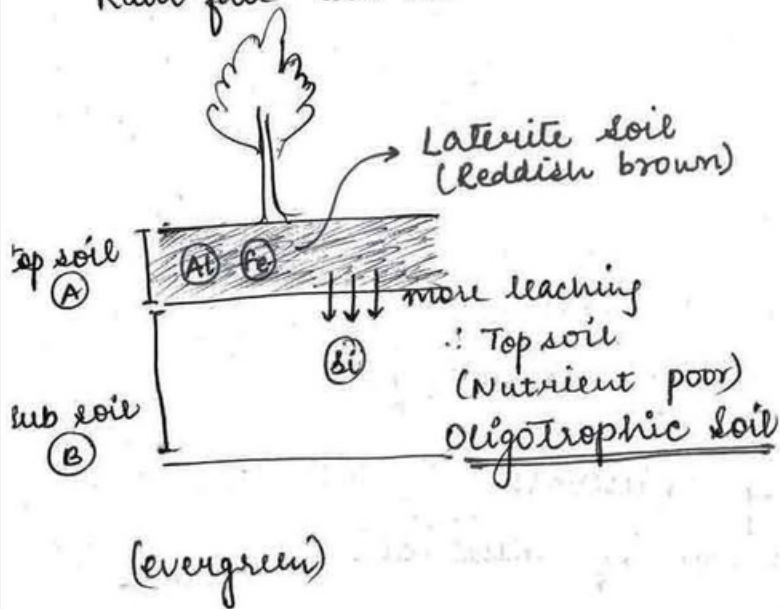
Pedology: study of soil (Edaphology)

eluviation: leaching of minerals

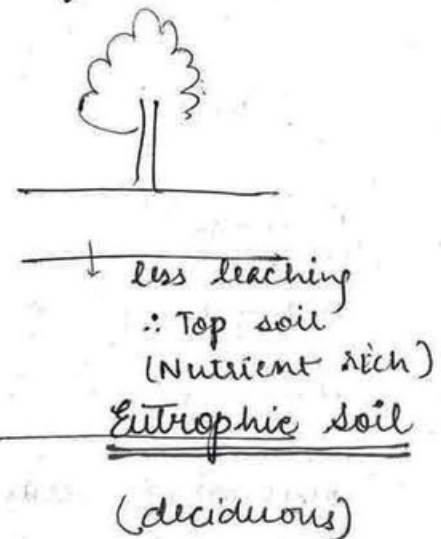
illuviation: accumulation of minerals.



Tropical ~~deciduous~~ rain forest
Rain fall - 200 cm



Temperate deciduous forest
Rain fall - 100 cm

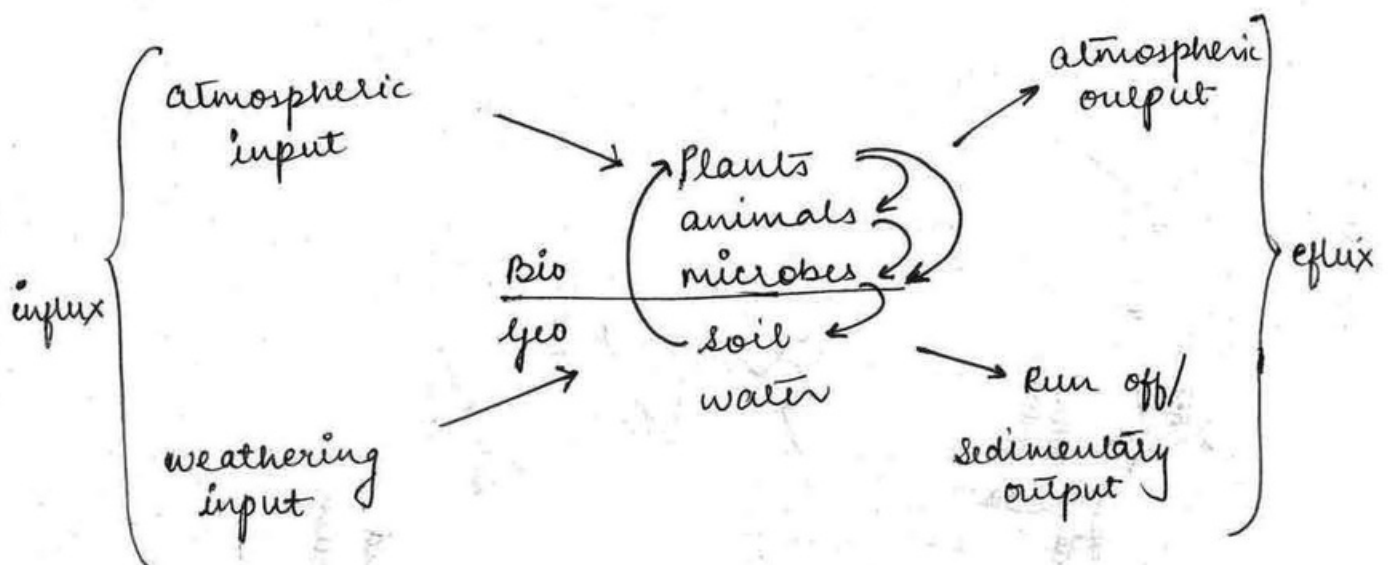
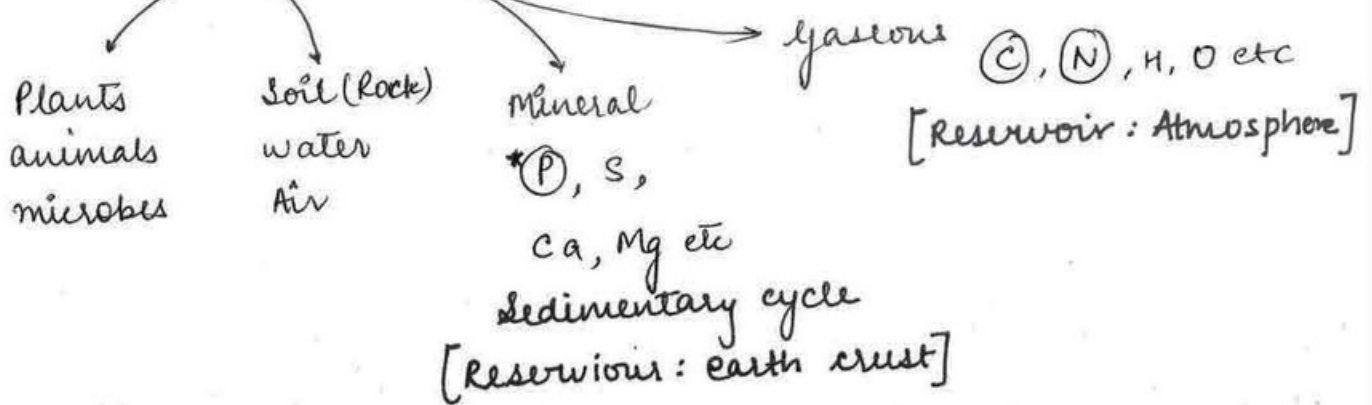


Salinity ↑ due to loggare of H₂O
for longer time

④ water ②

- » next to temp. (2nd most imp)
- » aquatic animals too face water - related prob.
(pH control : Euryhaline)
- » usually measured in ppt (parts per thousand)
- » { Inland water : less than 5%
sea water : 30-35%
hypersaline lagoons : more than 100% } } Saline %

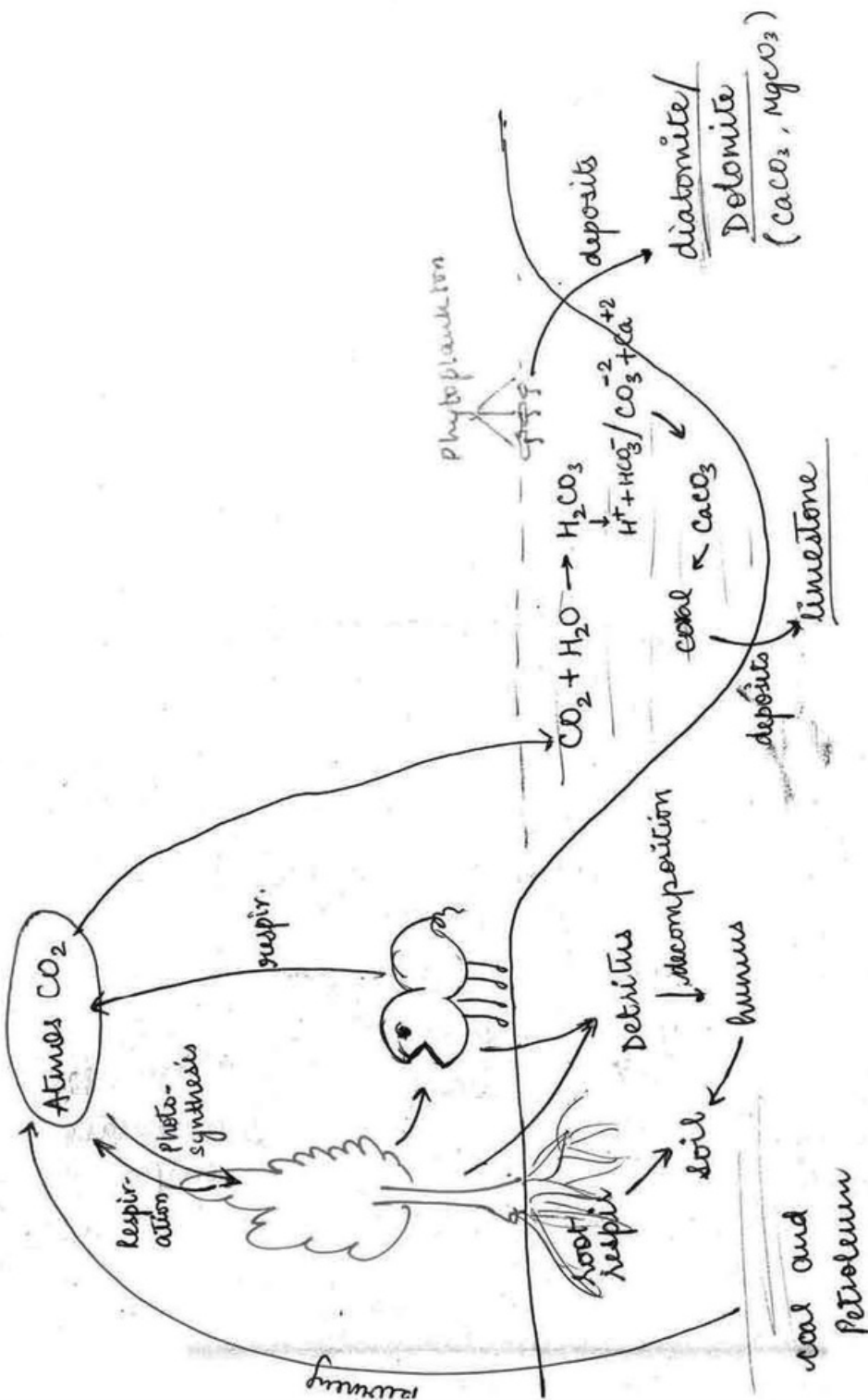
BIOGEOCHEMICAL CYCLE :



Carbon cycle :

Reservoirs:

- Atmosphere - CO_2
- lithosphere - Coal, Petroleum
- hydrosphere - carbonate (ocean) 71% of C

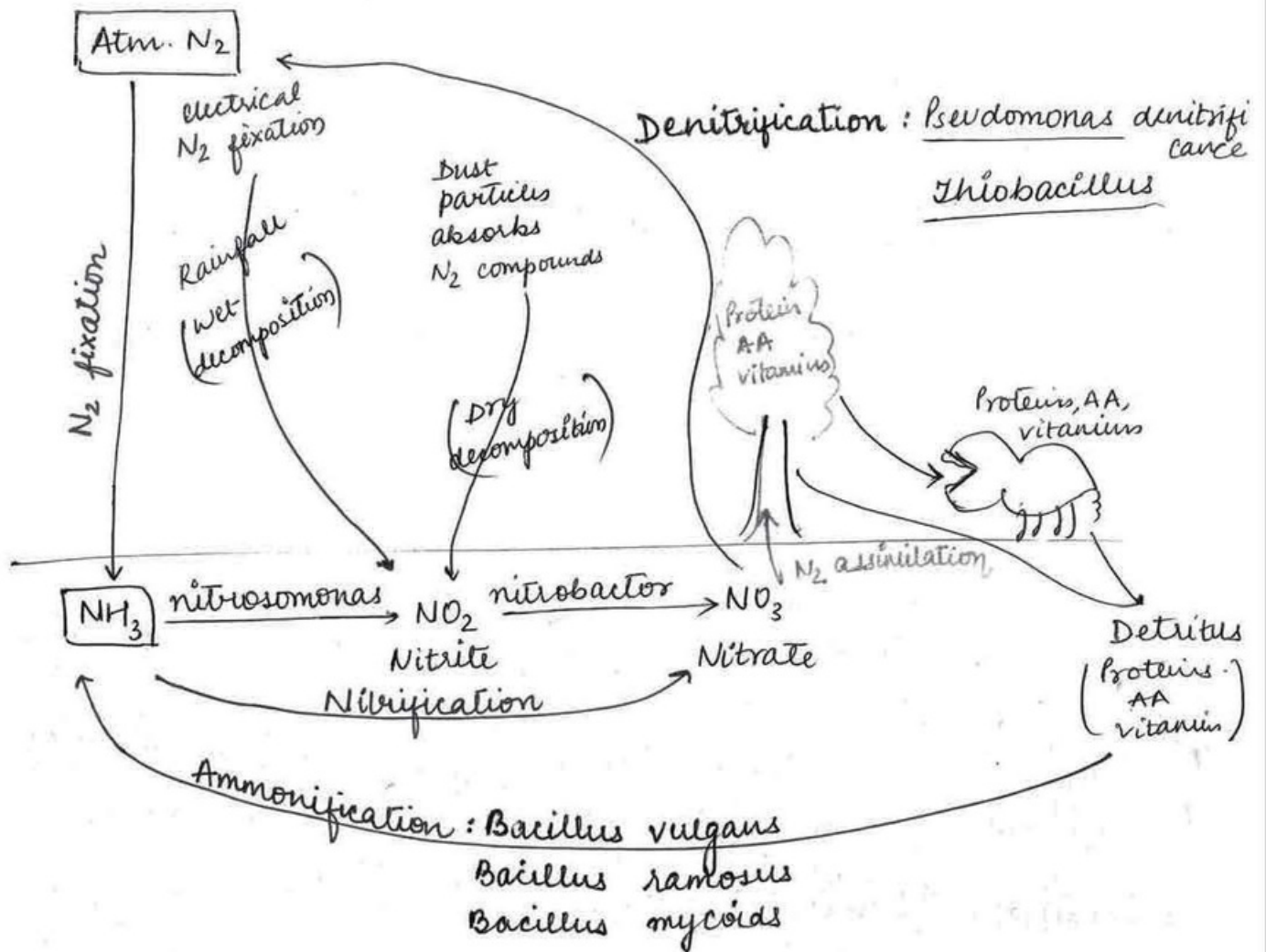


Annual photosynthesis carbon fixation ⇒ 4×10^{12} Kg

Annual photosynthesis

N_2 Cycle:

Reservoir = Atmosphere



*** N_2 fixation:

Symbiotic =

Rhizobium
Acetorhizobium
Azospirillum (woose symbiosis)

Frankia

Free living

Azotobacter = aerobic
Beijerinckia
Rhodospirillum

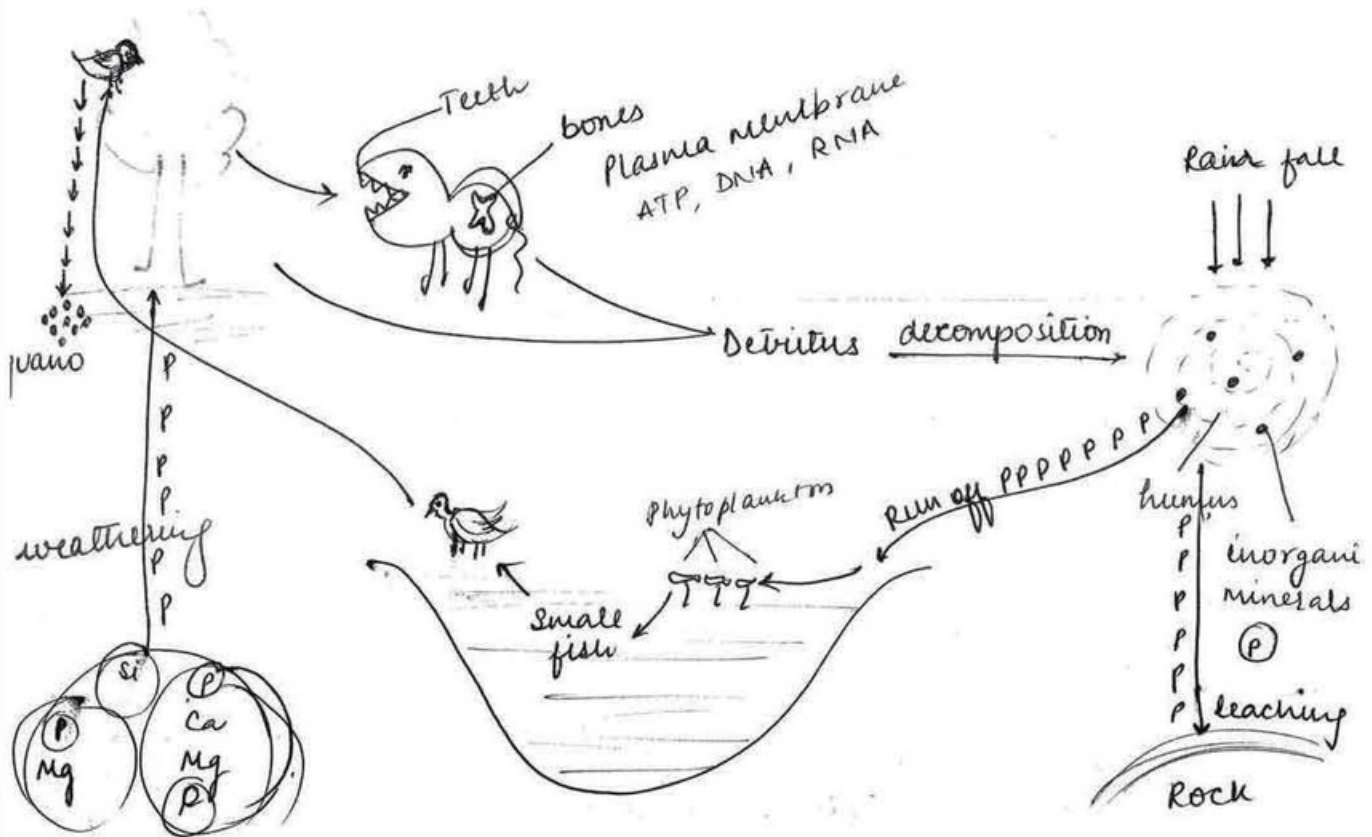
BCA

Nostoc
Anabaena } free living + symbiotic
Anulosia (free living)

Phosphorus cycle (sedimentary cycle)

Reservoir : (Rock = earth crust)

* Nutrient immobilisation helps in mineral conservation



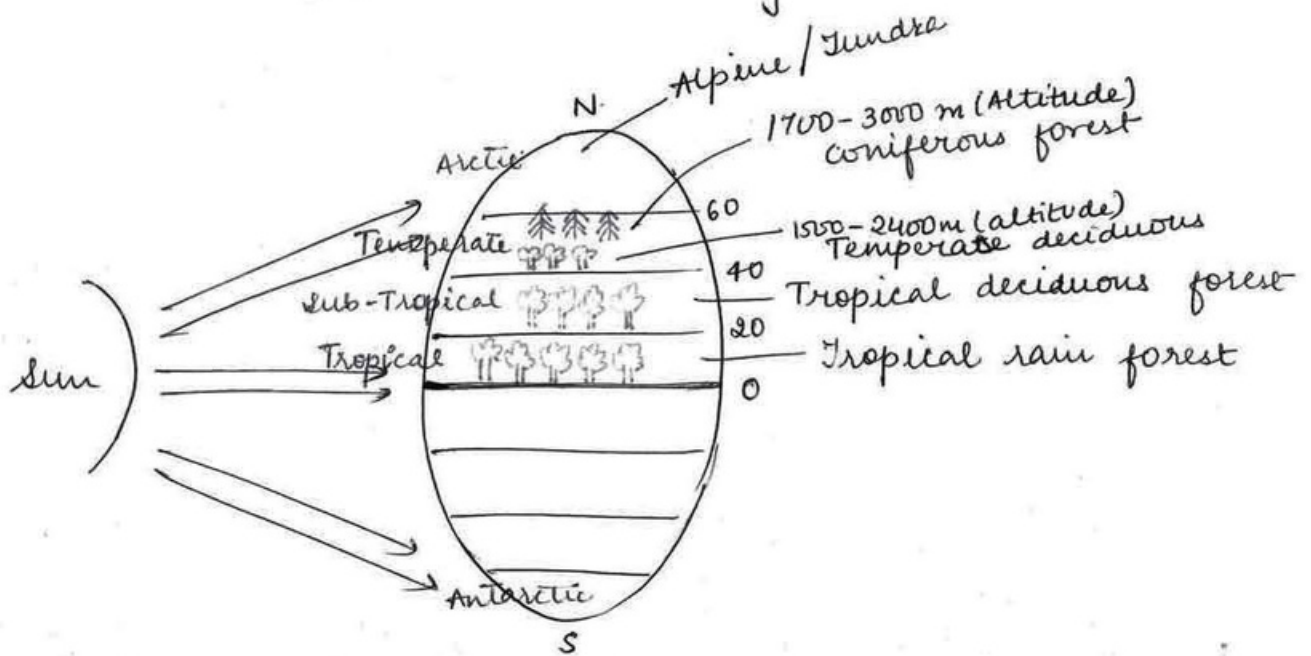
ecosystem services :

- > Purify air and water (wetland)
- > Minimise flood and drought
- > cycling nutrients
- > generate ~~to~~ fertile soil
- > Provide wildlife habitat
- > Maintain biodiversity
- > Pollinate crop
- > Storage of Carbon

Biome

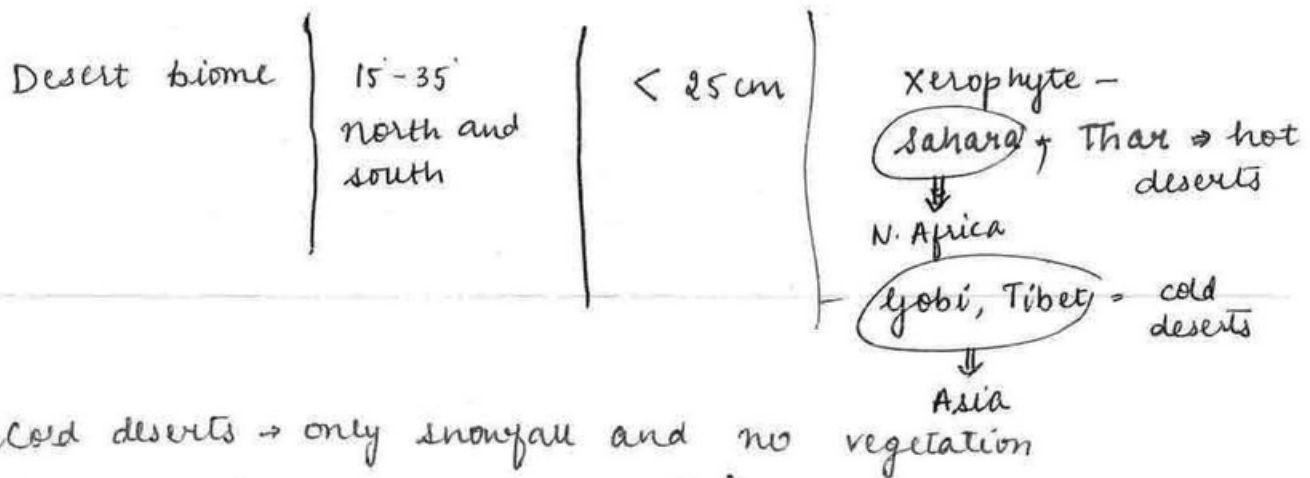
large ecosystem

- Formation :
- ① Temperature
 - ② Precipitation ← Rainfall
snowfall
 - ③ climax community



<u>Bio</u>	<u>Position</u>	<u>Rainfall</u>	<u>Vegetation</u>	<u>Special</u>
Tropical rain forest	0°-20° (equatorial region)	200-300 cm (2-3 months)	Hepes and Dipterocarpus	> Rich in flora and fauna > India: <u>eastern Himalaya</u> and <u>western Ghats</u> [Kerala, WB, Assam] > Diversity ↑, Rainfall ↑ Productivity ↑
Tropical deciduous forest	20°-40°	90-160 cm	Sal, Teak, Tundia	> Broad leaved forest > <u>major biome of India</u> > Plant shed their leaves before summer
Temperate Deciduous forest / <u>Oak forest</u>	40°-60° altitude: (1500-2400)m	100-250 cm	<u>Oak</u> (<u>Quercus</u>)	> Broad leaved forest > Plants shed their leaves in autumn and bear again in spring. > Dying autumn leaves become yellow, orange, R
coniferous	40°-60° altitude: (1700-3000)m	50-170 cm	Pinus Deodar Ahois	> Also called <u>Taiga</u> or <u>North wood</u> > Temperate <u>evergreen forest</u>

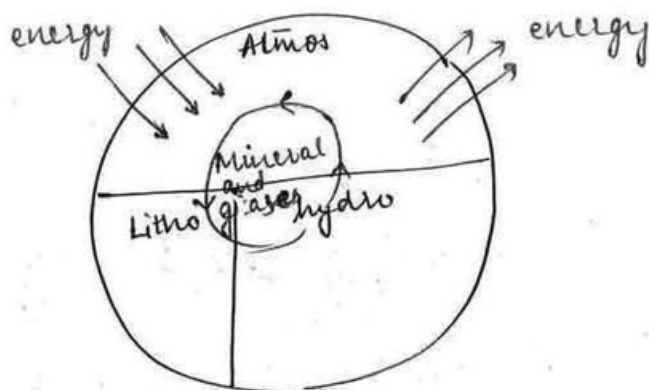
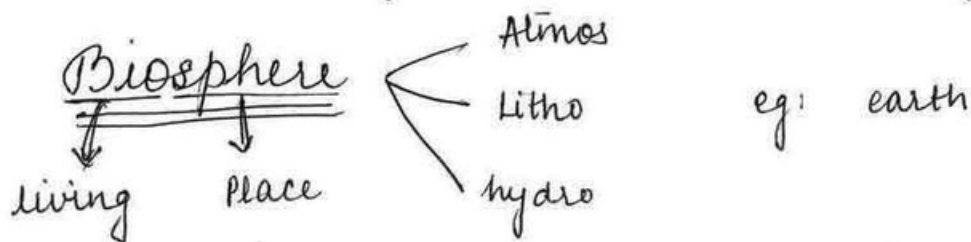
Tundra/ Alpine desert/ Arctic	Above 60°	25 cm	- Lichen moss - Rhododendron (angiosperm plant)	> also called parma frost (permanent snow) > food web simple > Delicate or fragile biome
Chaparral/ Mediterranean scrub (= shrub) forest	Pacific coast of North America and south coast of Australia	very less during winter	Drought and fire resistance plants	> mostly shrub are found
Tropical savanna Biome	South America South Africa North Australia North and East central India	Much less that does not support forest form but higher than desert	grass with scattered leaves (trees)	> Seasonal rainfall/high > hooped mammals are found
Grassland (temperate)	Prairies - N. America Pampas - S. America Steppes - Russia in Europe Tussock - New Zealand Veldts - Africa	25-75 cm	grass	> Rainfall during summer



Cold deserts → only snowfall and no vegetation

(Tundra - lichen and mosses ⊕)

(cold desert - only that area where no vegetation ⊕)



for energy transport (open biosphere)

for minerals and gases (close biosphere)

BIODIVERSITY

Acc. to ~~to~~ Edward Wilson

Biodiversity : Totality of gene, species and community

i) gene diversity

⊕nce of diff. types of gene in a species.

eg: Rauwolfia $\Rightarrow$ Reserpine

Rice $\Rightarrow$ > 50,000 strain

mango $\Rightarrow$ 1000 ~~var~~ varieties (= subspecies)

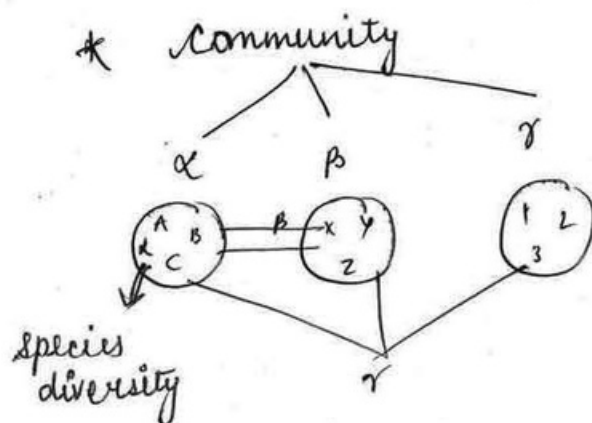
ii) species diversity

diff. species in a community

iii) community diversity / Ecosystem diversity ^(N)

↓
diff. community
in a ecosystem

↓
diff ecosystem
in a landscape



α diversity = within a community

β diversity = b/w 2 community

γ diversity = more than 2 community
(= in a ecosystem)

IUCN (2004)



1.5 million species — discovered

22%

> 70% animals $\frac{7}{10}$ (insects)
Max - insects

Robert May $\Rightarrow$ 7 Million
(Total estimated value)

> 22% Plants

Max - Angio > fungi

Prokaryotes are not included :

conventional / Traditional method can't be used
and even many of them can't ~~even~~ be cultured.

India :

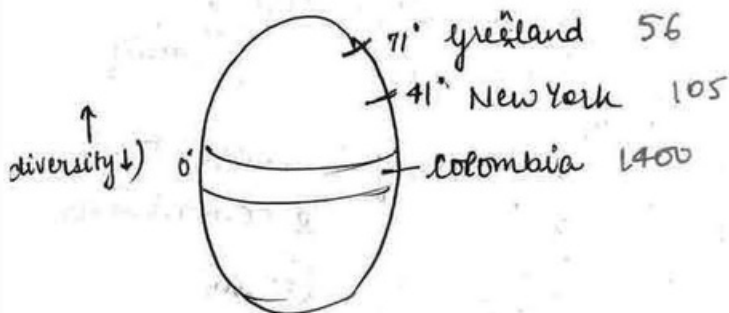
world $\Rightarrow$ 2.4 % of ^{land} earth area

Diversity $\Rightarrow$ 8.1 %

(12 mega diversity)

Pattern of Biodiversity :

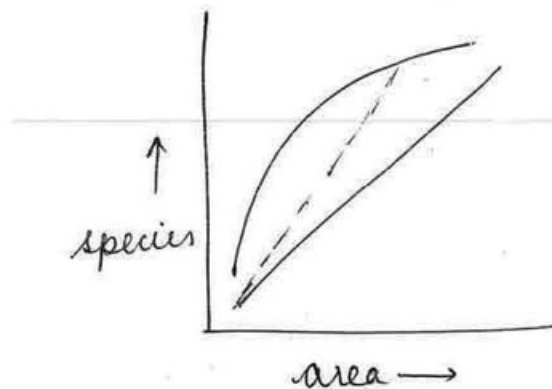
i) latitudinal gradient



Amazon forest : lungs of earth
(South America) 20% of total earth atmosphere

ii) species area relationship

area $\uparrow$, species $\uparrow$ upto a limit



$$S = CA^Z$$

$$\log S = \log C + Z \log A$$

$\downarrow$ species $\downarrow$ Y-intercept $\downarrow$ slope $\downarrow$ area

$$Z = 0.1 - 0.2$$

$$Z = 0.6 - 1.2$$

$$Z = 1.15 \text{ (tropical dense area) } \neq \text{ fruit eating bird}$$

characters:

analytical $\Rightarrow$ 1st actual data

synthetic $\Rightarrow$ other assumption acc to graph

- a) critically endangered : immediate future in danger
- b) endangered : near future in danger
- c) vulnerable : more in no. but under over exploitation
- d) rare : least in no in nature

Pollution

Pollutants :

- ① Non-biodegradable :
- | | |
|--------|------------|
| DDT |] compound |
| Hg | |
| Al | |
| phenol | |

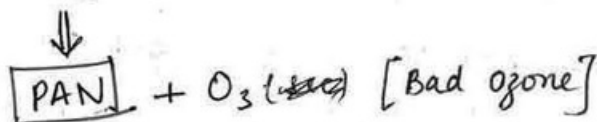
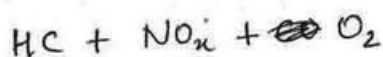
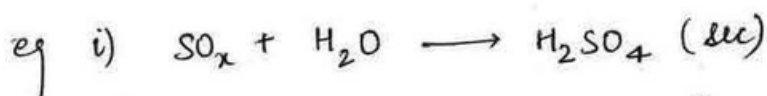
- ② Biodegradable :
high quantity $\Rightarrow$ then pollutant

- ③ Primary Pollutant

vehicular smoke $\Rightarrow$ $[CO_2, CO, NO_x, SO_x, Pb, HC]$, DDT

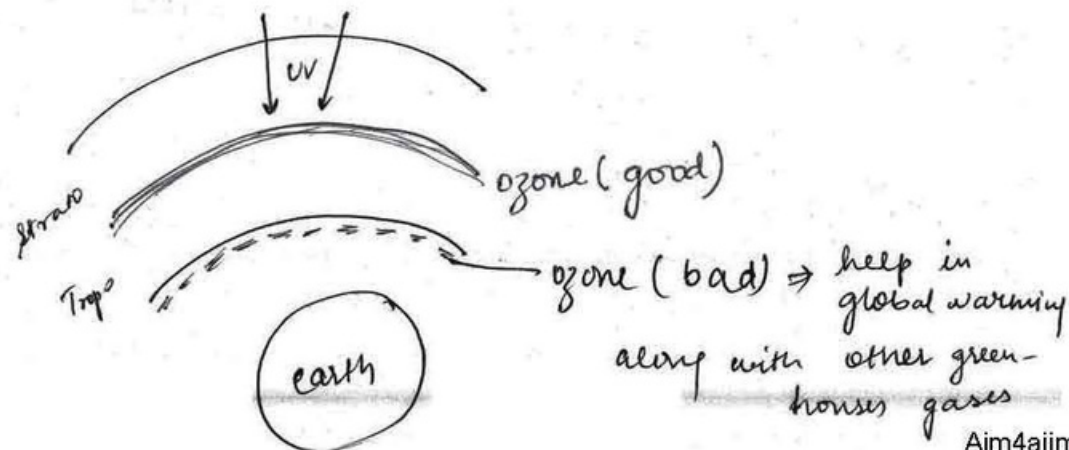
- ④ Secondary Pollutant

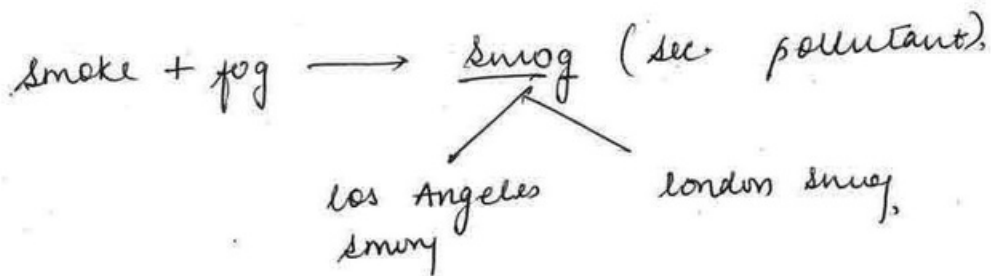
Primary + Primary = secondary, [SYNERGISM]



(Peroxyacetyl Nitrate)

both are sec pollutants



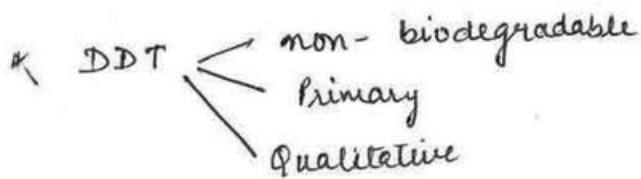


⑤ Quantitative

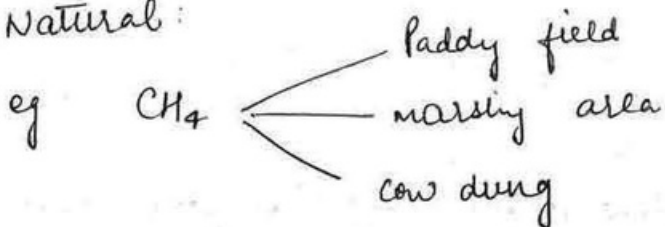
- > Part of nature
- > Quantity less : causes pollution
- > eg: CO_2 , NO_2

⑥ Qualitative

- > not a part of nature
- > man-made
- > small quantity - causes pollution
- > eg: CO, CN, DDT



⑦ Natural:

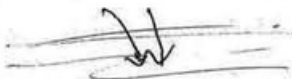


Methanogens

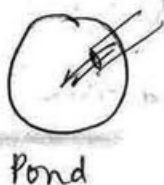


Positive Pollution (by adding)

- > over fertilizers
- > land filling



Point source
pollute small area



Negative Pollution (by removing)

- > Overgrazing
- > Soil erosion



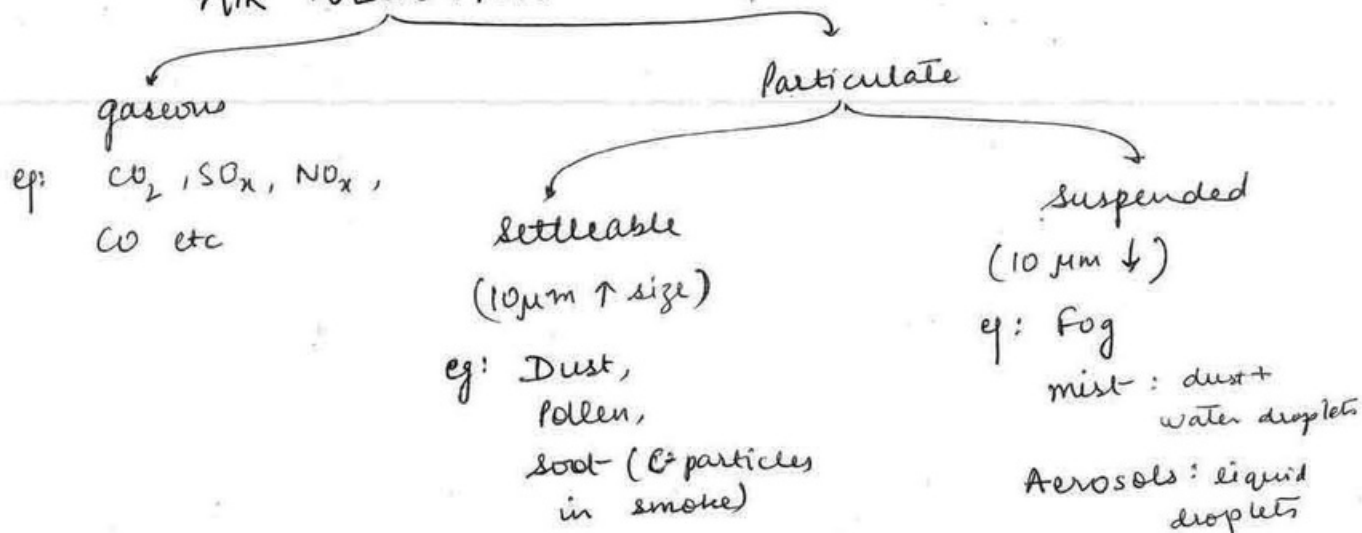
Non-point source
pollute large area



⇒ fix source

mobile source

AIR POLLUTION :



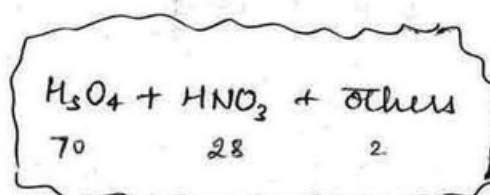
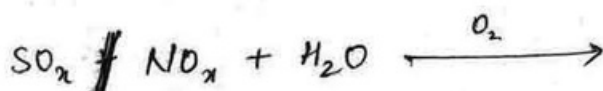
Major Primary Air Pollutants

① NO_x : sources - vehicles
Agricultural land
agriculture industries.

② SO_x : source - coal burning
(thermal power plant) $\leftarrow$ Air
oil refineries $\leftarrow$ water
 $\leftarrow$ soil

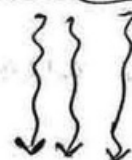
Secondary air pollutant

① Acid Rain:

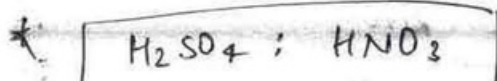


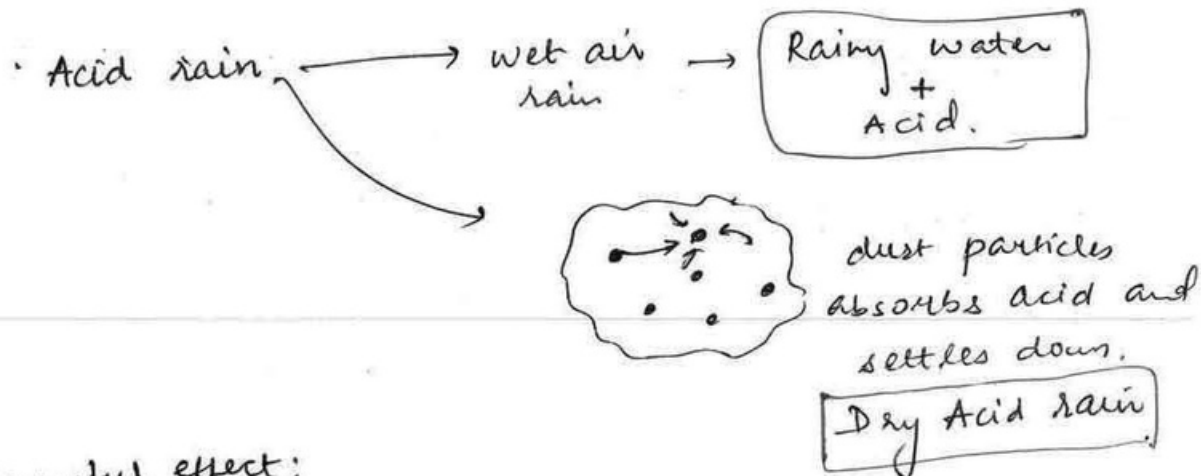
Rainy water $\text{pH} = 5.6$

Acid rain $\text{pH} < 5.6$



acid rain





Harmful effect:

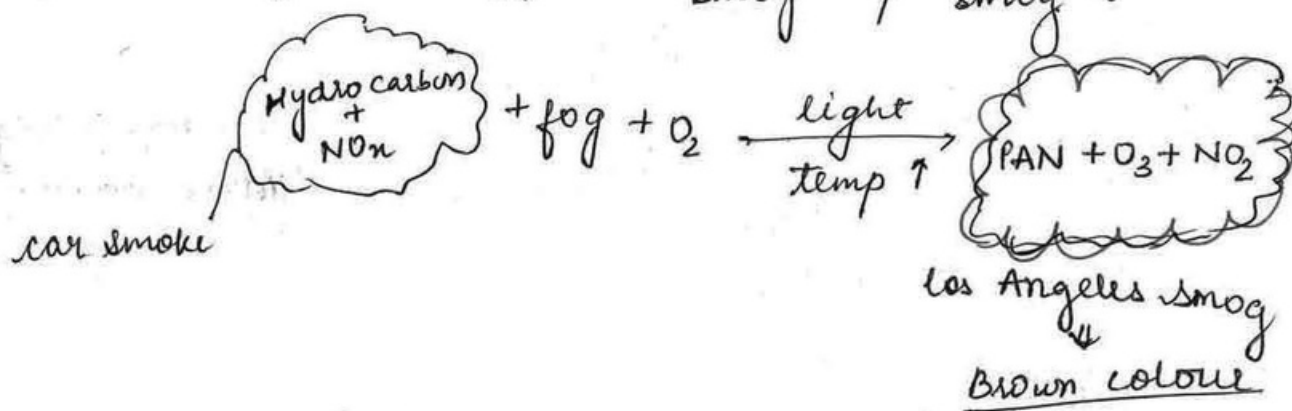
- ① Air, water and soil becomes acidic
- ② marble turns ~~yellowish~~ yellow in colour.

STONE LAPROSY

eg: Taj Mahal

② smoke + fog with smog

i) Los Angeles smog / photochemical smog / oxidising smog



low temp → incomplete smog

↓

grey colour

harmful effects of PAN :

① eyes, skin irritation

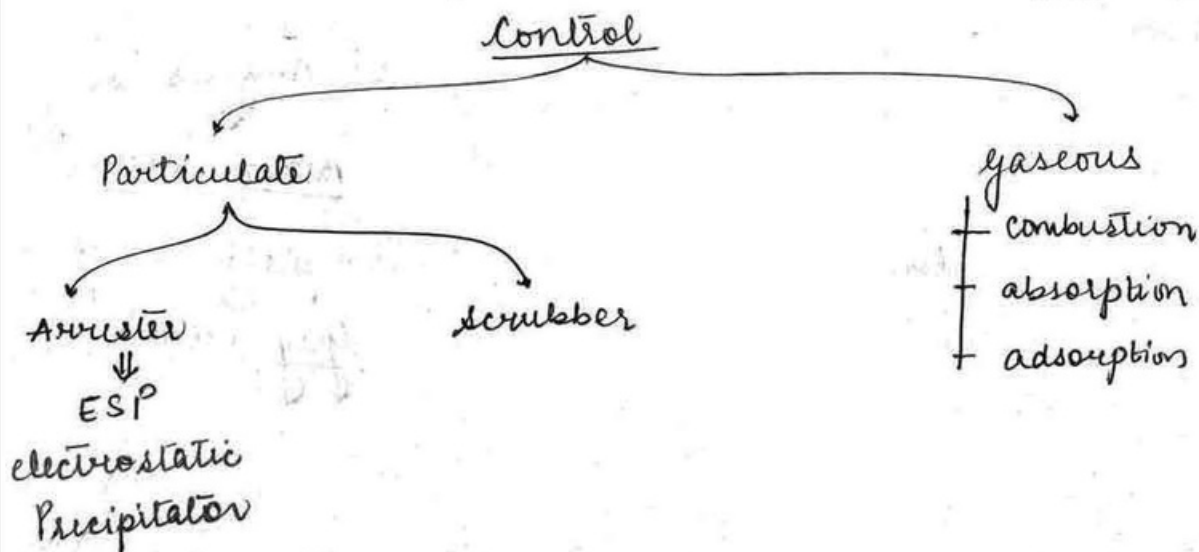
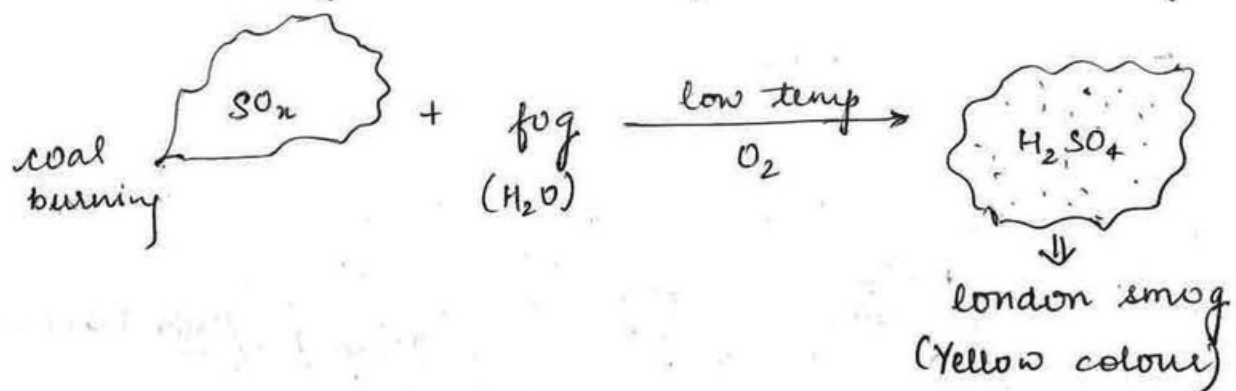
② Respiration blockage

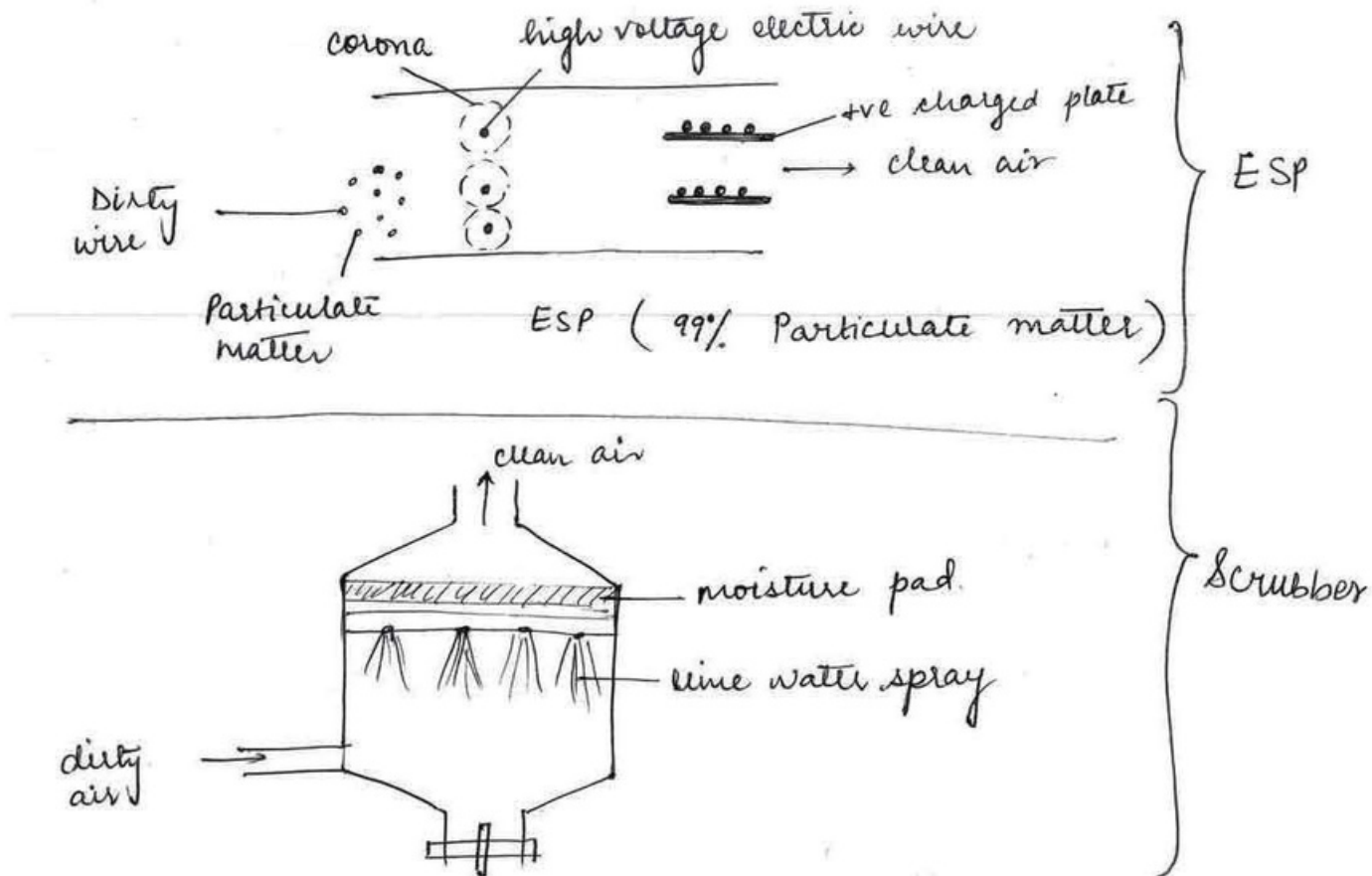
③ Plants $\begin{cases} \text{PS II } e^- \text{ flow block} \\ \text{chlorophyll syn. block} \end{cases}$

Ozone ① emphysema

② damage mucus membrane of respiratory tract

ii) London smog / classical smog / coal induced smog :





Euro norms :

- ① Euro - I $\Rightarrow$ June 1999
- ② Euro II / B.S II $\Rightarrow$ 1 April 2000
 ↓ Bharat stage ↓
 11 major polluted cities
- ③ Euro III / B.S III $\Rightarrow$ April 2005
 ↓
 11 major cities Rest of India
 ↓
 Euro II
- ④ Euro IV / B.S IV $\Rightarrow$ April 2010
 11 major cities + Rest of India
 2 cities ↓
 Euro III

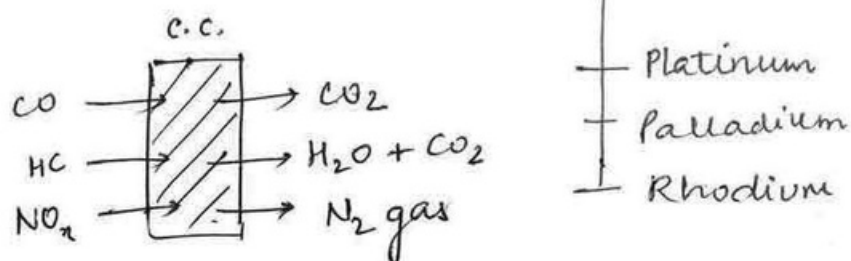
Euro II - Sulphur
 Diesel - 350 ppm
 Petrol - 150 ppm
Hydrocarbon
 42%

Euro II / IV - Diesel
Petrol] - 50 ppm
 35%

Engine :

① Multipoint fuel injection
 ↓
 for efficient burning of hydrocarbons

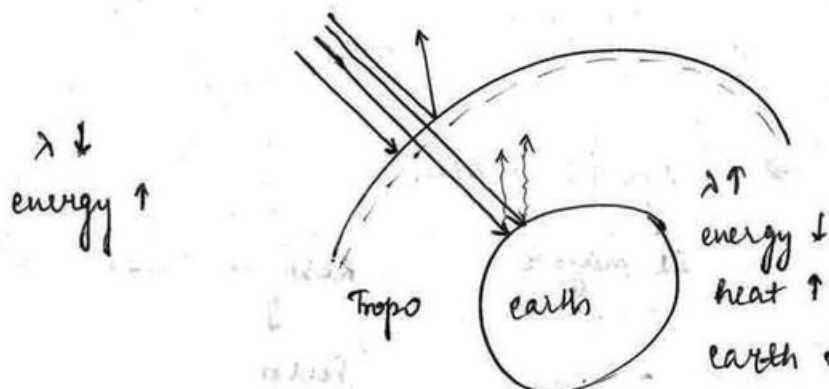
② Catalytic converter :



~~Lead free~~ : grey petrol (unleaded petrol)

green house effect :

230 ppm → 368 ppm → 394 ppm
 CO₂ - 60%
 CH₄ - 20%
 CFC - 14%
 N₂O - 6%
 } radio-actively active gases

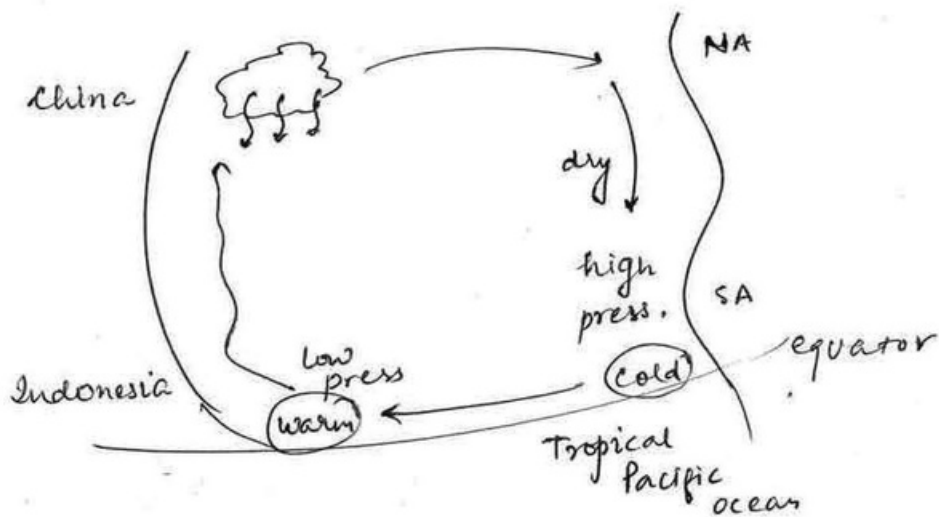


Earth surface becomes WARMER
 GLOBAL WARMING

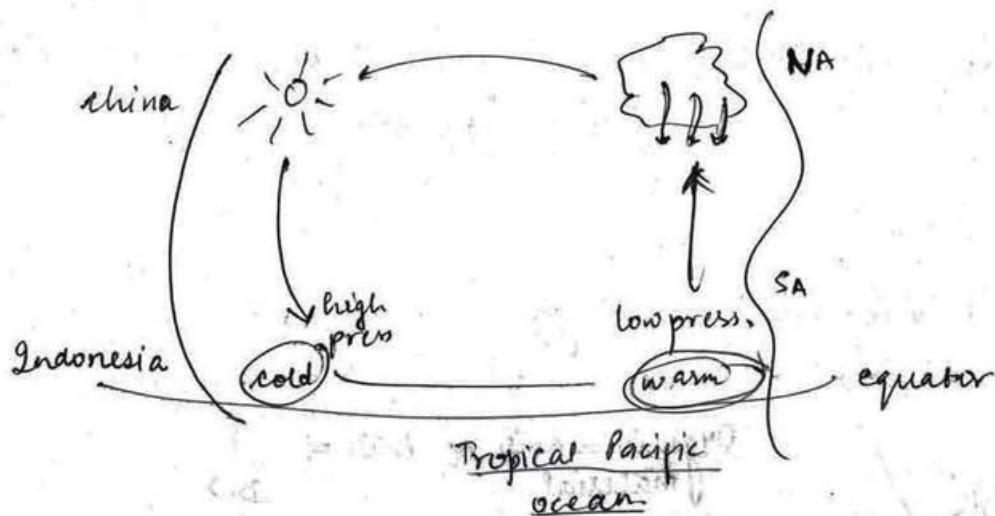
CO_2 fertilizing effect :

$\text{CO}_2 \uparrow \rightarrow \text{C}_3 \text{ plants} \rightarrow \text{Photosynthesis} \uparrow$
Productivity $\uparrow$
Partial closing of stomata
Transpiration $\downarrow$

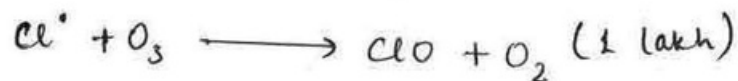
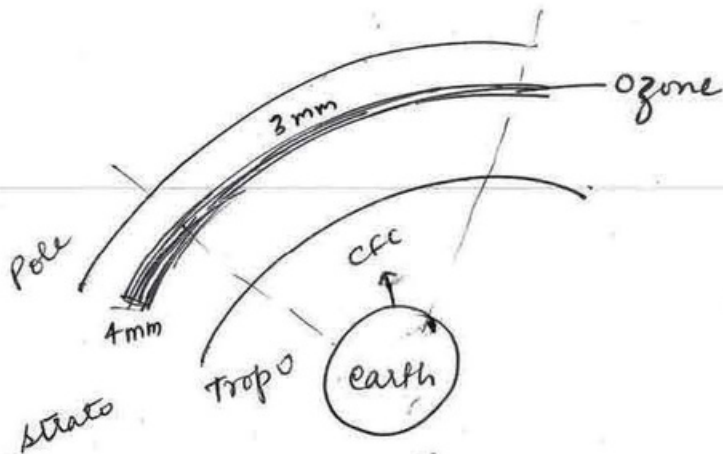
La-nina



El nino



Ozone Depletion



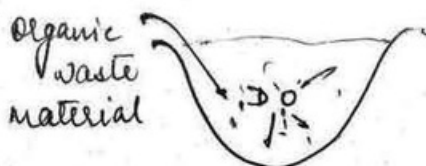
* Kyoto Protocol (Japan) COP = 3

1 active Cl can break 1 lakh ozone

Ozone depleting substances : Montreal Protocol (Canada)
1987 (effective: 1989)

WATER POLLUTION

① BOD : Biochemical oxygen demand

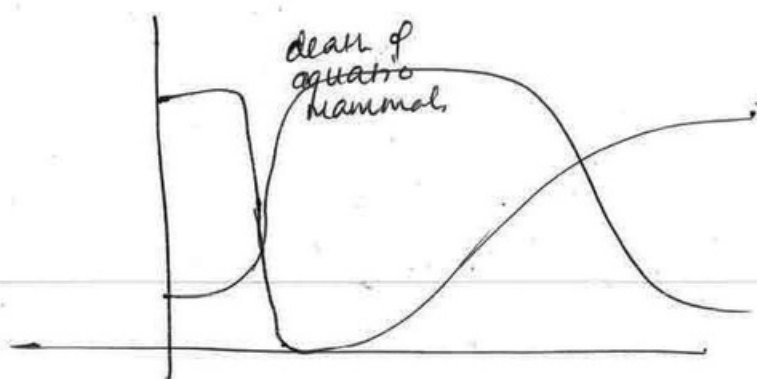


Organic waste material $\propto$ BOD $\propto \frac{1}{\text{DO}}$

dissolved oxygen $< 8 \text{ mgL}^{-1} \rightarrow$ Polluted

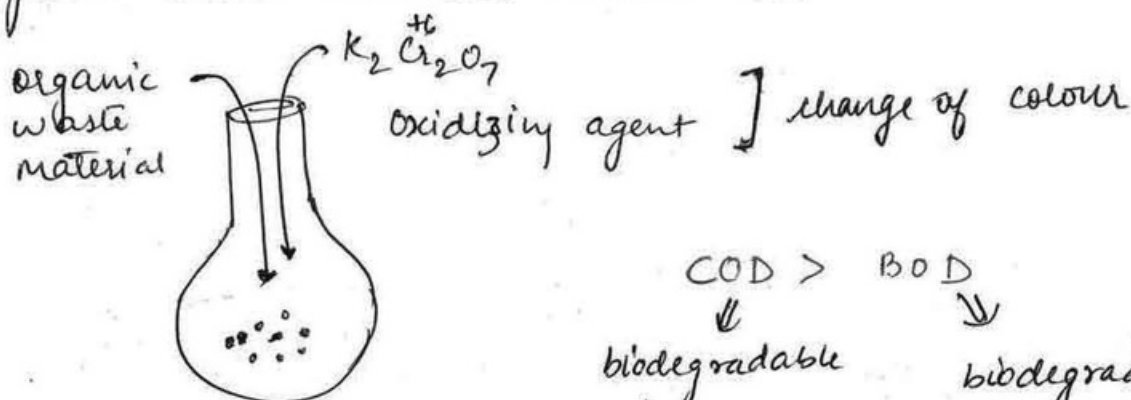
$\text{DO} < 4 \text{ mgL}^{-1} \rightarrow$ heavy polluted

844 meter means DO



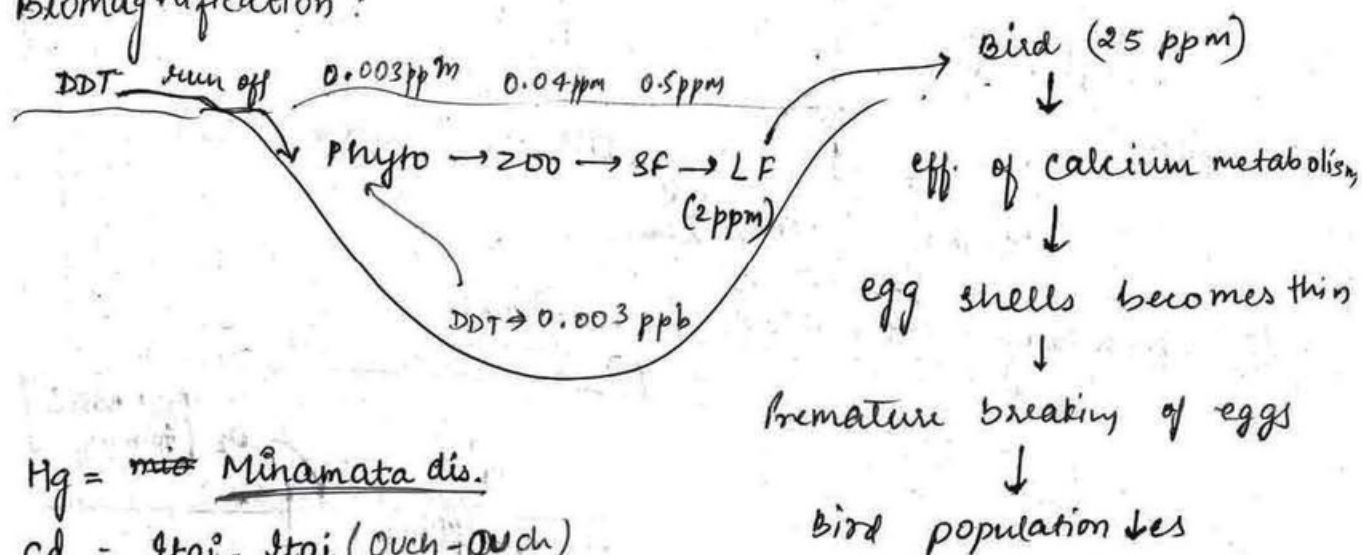
① COD (chemical oxygen demand)

Oxygen demand of chemical for decomposition of organic waste material called COD



COD > BOD
 ↓ ↓
 biodegradable + non biodegradable

Biomagnification:

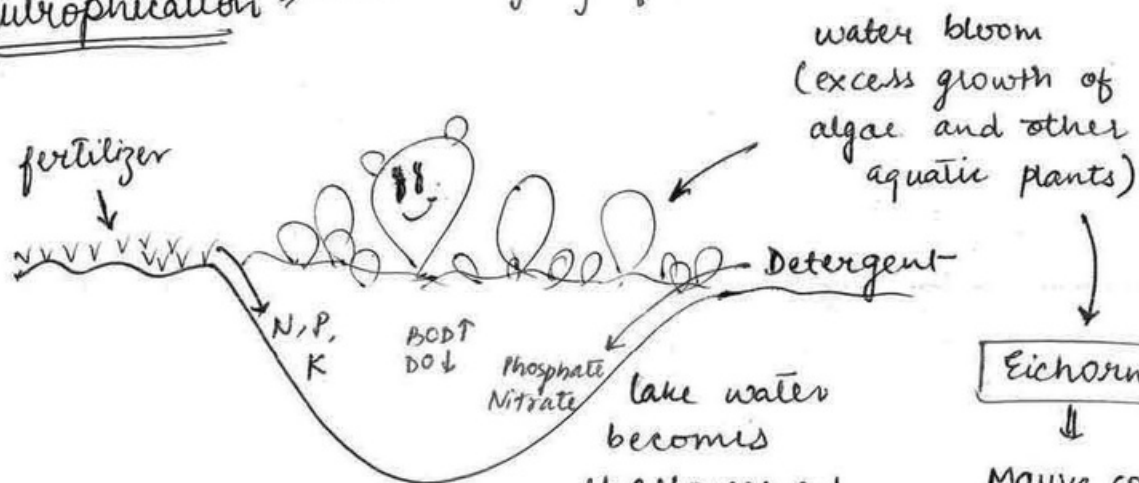


Hg = ~~mis~~ Minamata dis.

Cd = Itai-Itai (ouch-ouch)

PCB ⇒ Polychlorinated Biphenyls (Max Biomagnification)

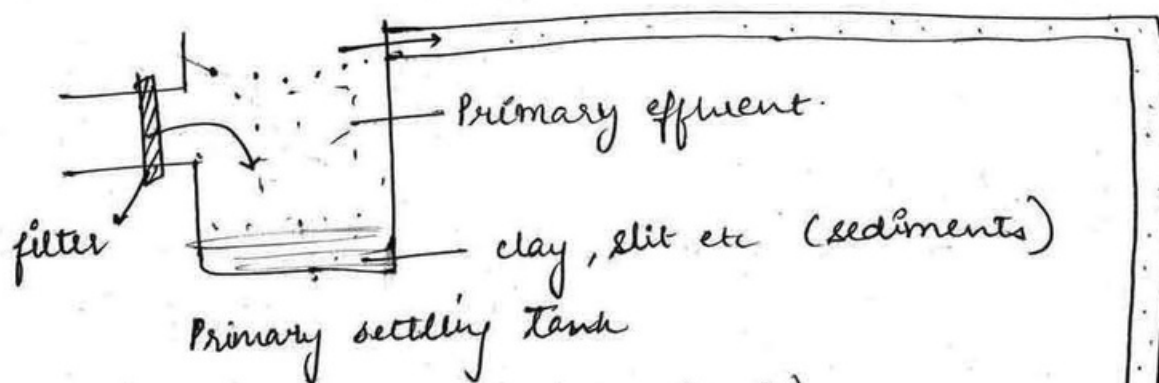
Eutrophication $\Rightarrow$ Natural ageing of lake



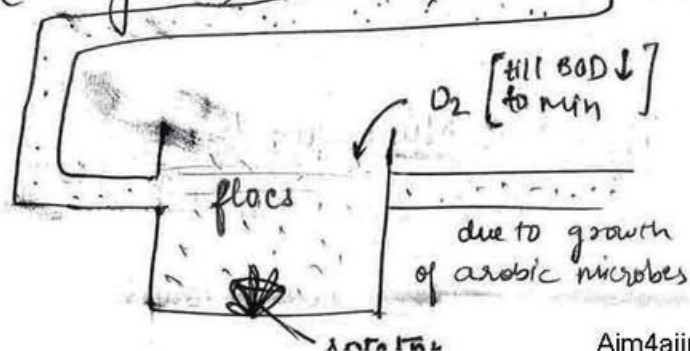
eutrophication, due to human activities called accelerated/cultural eutrophication

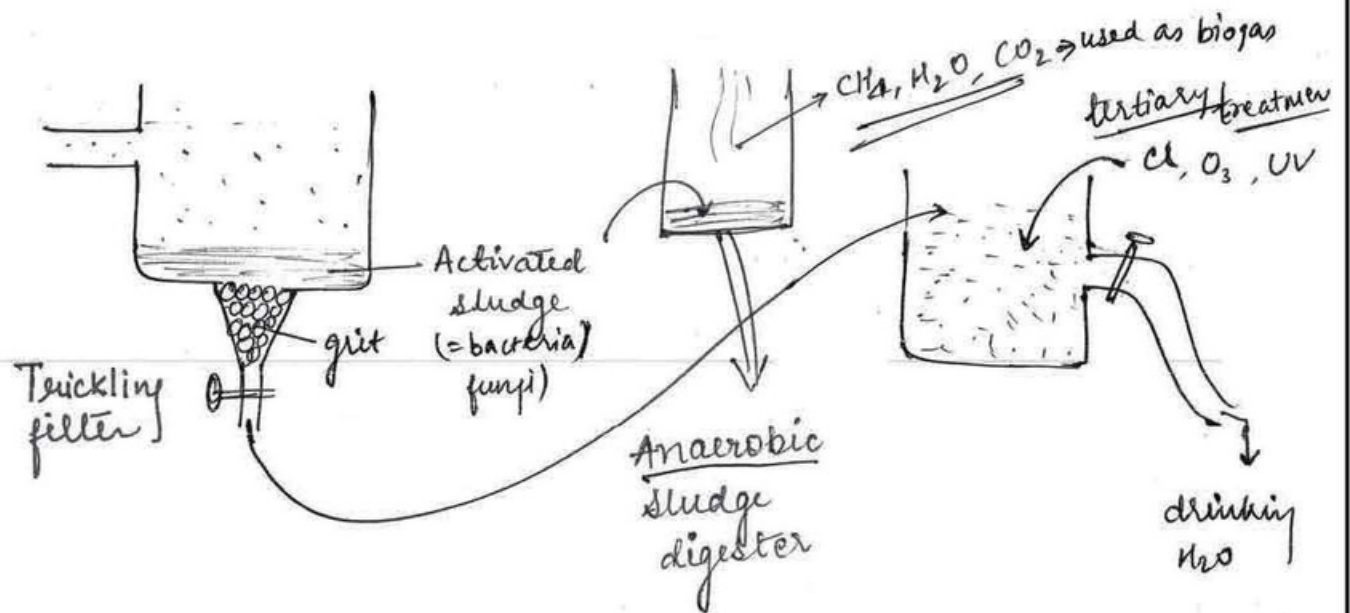
Sewage Treatment

- ① Primary treatment (Physical)
 - i) filtration
 - ii) sedimentation



- ② Secondary treatment (Biological)





- i) Aeration tank
- ii) Secondary settling tank / trickling filter
- iii) Anaerobic sludge digester